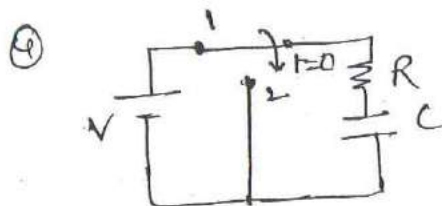
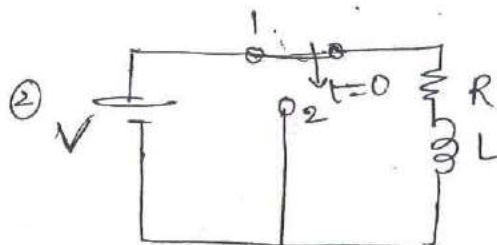
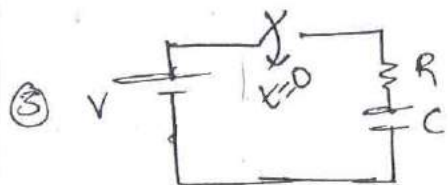
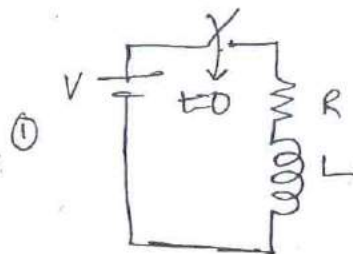
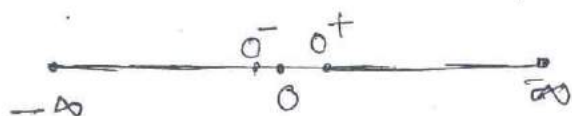


# TRANSIENTS



For ckt ①:  $t=0^-$ ,  $i=0 \rightarrow 0 \text{ i.c}$

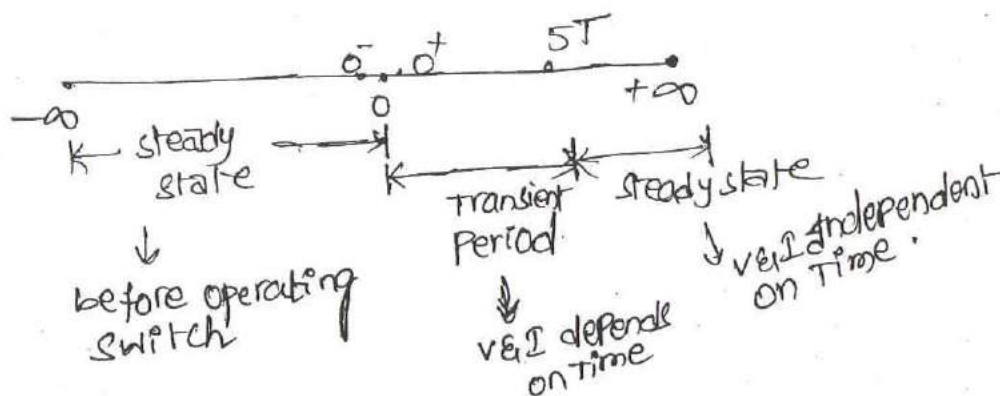
$t=0^+$ ,  $i=0 \rightarrow \text{that means } 0 \text{ i.c}$

$t=\infty$ ,  $v_L = L \frac{di}{dt} = 0$  since at  $t=\infty$ , current is constant  $\frac{di}{dt} = 0$

(S.C)  $\Rightarrow \boxed{i = V/R}$

conclusion:-

So the inductor behaviour changes w.r.to time.



For ckt ②:

$t=0^-$ ;  $i = I_0$

$t=0^+$ ;  $i = I_0 \rightarrow$  so inductor acts as a current source.

$t=\infty$ ;  $i = 0 \rightarrow$  since no energy present at inductor.

ckt-③ :-

$$t=0^- \rightarrow V_C = 0$$

$$t=0^+ \rightarrow V_C = 0 \rightarrow i.e. (s.c)$$

$$t=\infty \rightarrow V_C = V \rightarrow (o.c)$$

$$i = \frac{V - V_C}{R} = 0 \dots$$

capacitor behaviour changes w.r. to Time.

for ckt ④ :-

$$t=0^-, V_C = V_0 \rightarrow \text{Initial voltage}$$

$$t=0^+, V_C = V_0 \quad (\text{Capacitor acts as a voltage source})$$

$$t=\infty; V_C = 0 \rightarrow \text{Total energy will be discharged so } V_C = 0$$

Transients are present in the N/w when N/w is subjected to any changes either by changing source magnitude (or) by changing any circuit elements provided circuit consists of any energy storage elements.

Transients are present in the N/w when N/w is having any energy storage elements. Since 1. Inductor does not allow sudden change of current & Inductor stores energy in the form of magnetic field.

2. Capacitor does not allow sudden change of voltage & it stores energy in the form of electric field.

When N/w is having only resistive elements no transients are present in the N/w. Since -

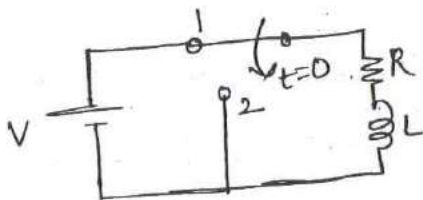
1. Resistor allows sudden change of current & sudden change of voltage.

2. Resistor does not store any energy.

Note:

1.  $t=0^-$  Indicates immediately before operating switch.
2.  $t=0^+$  Indicates immediately after operating switch.
3.  $t=\infty$  Indicates steady state condition.

Source free RL circuit :-



$$t=0^-; i = i_0$$

$$t=0^+; i = i_0$$

Corresponding to  $t \geq 0$  source is disconnected from ckt. so this is called "source free ckt".



KVL  $iR + L \frac{di}{dt} = 0$

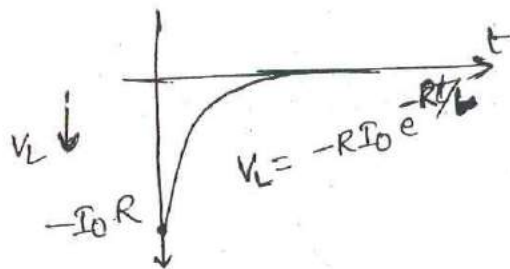
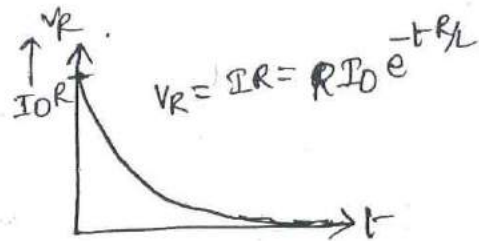
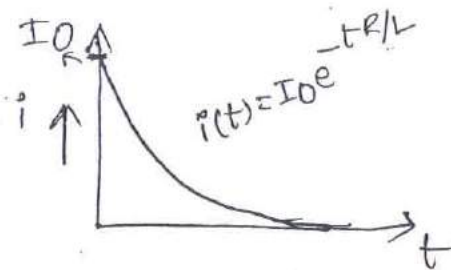
$iR = -L \frac{di}{dt}$

$-\frac{R}{L} i = \frac{di}{dt}$

$\Rightarrow \int_0^t -\frac{R}{L} dt = \int_{I_0}^{i(t)} \frac{di}{i}$

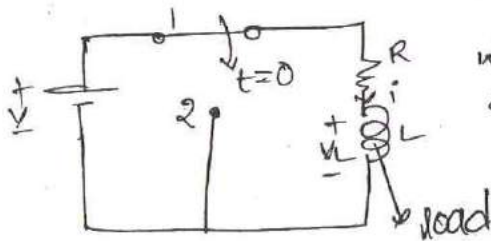
At  $t=0$ ;  $i = I_0$  so  
 $t=t$ ;  $i = i(t)$  so upper limit  $= i(t)$

$\Rightarrow i(t) = I_0 e^{-\frac{R}{L}t}$   
 $i(t) = I_0 e^{-t/\tau}$



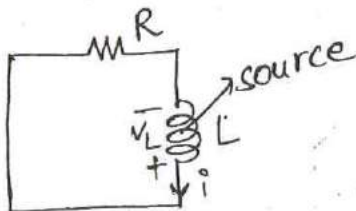
$V_L = L \frac{di}{dt}$   
 $= L \cdot \frac{d}{dt} (I_0 e^{-\frac{R}{L}t})$   
 $V_L = -I_0 \cdot R e^{-\frac{R}{L}t}$

now take the given ckt.



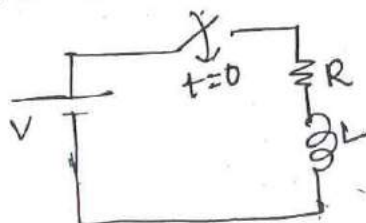
when switch is at position 1, then inductor acts as a load.

$t > 0$  :-



NOTE:- In the above ckt when switch is transferred from position ① to position ② inductor current direction does not change, but voltage across the inductor polarities are reversed.

RL circuit with source :-



- 1) classical method
  - 2) Laplace T/F method
- for objective ques.  $\downarrow$  for conventional paper

$\rightarrow$  we can able to observe the charging & discharging of L & C.

$$V = iR + L \frac{di}{dt}$$

$$\Rightarrow \frac{di}{dt} + \frac{R}{L}i = \frac{V}{L}$$

$$\boxed{\frac{di}{dt} + \frac{R}{L}i = \frac{V}{L} \quad //..}$$

$$i(t) = C.F + P.I$$

P.I  $\rightarrow$  steady state response  
(or) final value.

$\rightarrow$  Inductor acts as a short ckt under steady state condn.

$$\Rightarrow i = \frac{V}{R} = i(\infty)$$

$$\therefore \begin{cases} i(t) = C.F + P.I \\ i(t) = A e^{-\frac{R}{L}t} + \frac{V}{R} \end{cases}$$

At  $t=0^-$  ; Switch was open ;  $i=0$   
 $t=0^+$  ;  $i=0$   
 $t=\infty$  ;  $i = A + \frac{V}{R}$   
 ....

At  $t=0$   
 $\downarrow$   
 $i(0) = A(1) + \frac{V}{R}$   
 $0 = A + \frac{V}{R}$

$$\Rightarrow A = 0 - \frac{V}{R}$$

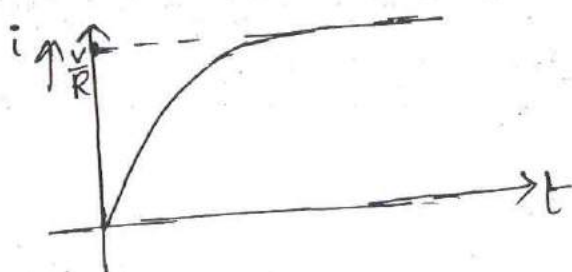
$$\boxed{A = i(0^+) - i(\infty)}$$

For RL & RC N/w this is the valid general expression.

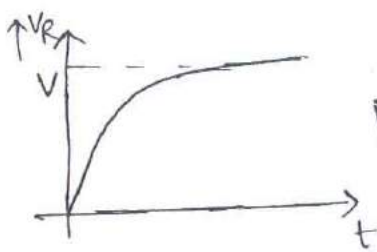
$$\boxed{i(t) = [i(0^+) - i(\infty)] e^{-\frac{R}{L}t} + i(\infty)}$$

This is General expression for RL ckt & RC ckt. Just  $\frac{R}{L}$  & RC are interchange

$i(0^+) \rightarrow$  Initial value  
 $i(\infty) \rightarrow$  final value



$$i(t) = \frac{V}{R} (1 - e^{-\frac{R}{L}t})$$



$$V_R = IR$$

$$V_R = V(1 - e^{-\frac{R}{L}t})$$

$$V_L = L \frac{di}{dt}$$

$$= L \frac{d}{dt} \left( \frac{V}{R} (1 - e^{-\frac{R}{L}t}) \right)$$

$$V_L = V \cdot e^{-\frac{R}{L}t}$$

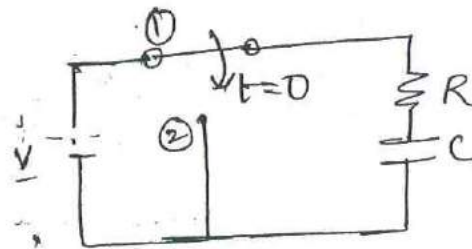


Note:-

from the above 3 graphs At  $t = \infty$ ,  
the Inductor voltage is zero i.e. short circuit  
and current is maximum.

Source free RC circuit:-

At  $t=0$  see the switch is transfer from position ① to position ②  
so source is disconnected.



$$I_R + I_C = 0$$

$$\frac{V}{R} + C \frac{dV}{dt} = 0$$

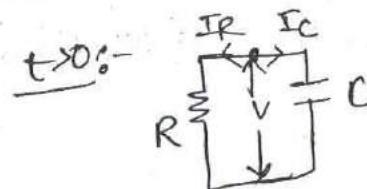
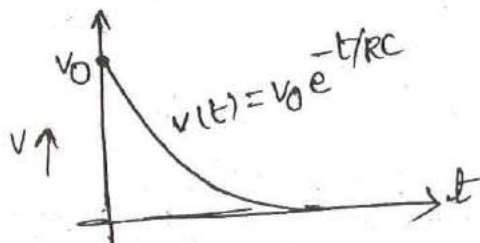
$$\frac{V}{R} = -C \frac{dV}{dt}$$

$$-\frac{1}{RC} dt = \frac{dV}{V}$$

$$\int_0^t \frac{1}{RC} dt = \int_{V_0}^{V(t)} \frac{dV}{V}$$

switching is starts from 0 time.  
so lower limit = 0.

$$\Rightarrow V(t) = V_0 e^{-t/RC}$$



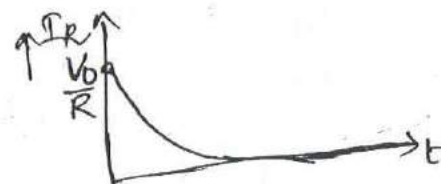
$$t = 0^-; V_C = V_0$$

$$t = 0^+; V_C = V_0$$

$$I_R = \frac{V}{R}$$

$$= \frac{V_0 e^{-t/RC}}{R}$$

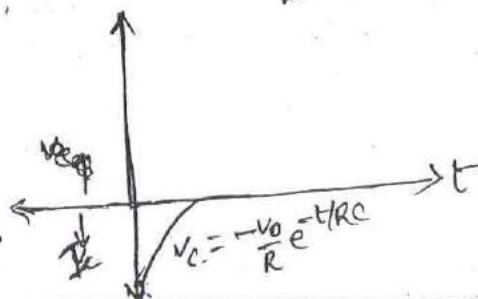
$$I_R = \frac{V_0}{R} e^{-t/RC}$$

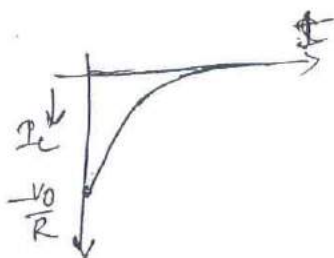


$$I_C = C \frac{dV}{dt}$$

$$= C \cdot \frac{d}{dt} (V_0 e^{-t/RC})$$

$$I_C = -\frac{V_0}{R} e^{-t/RC}$$





Note:-  
from position ① to ②  
voltage across the  
Capacitor does not  
change. but current  
is reversed.

$$I_R = I_C$$

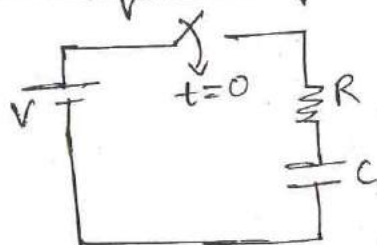
but at starting case we are consider  
 $I_C$  is downwards, so  $I_C$  becomes  $-ve$ .

$$I_R = -I_C$$

$$\Rightarrow I_R + I_C = 0$$

NOTE:- when switch is transferred from position ① to position ②  
voltage across the capacitor polarities does not change. but  
current direction of the capacitor is reversed.

RC circuit with Source :-



By KVL:-

$$V = V_R + \frac{1}{C} \int i dt$$

$$\frac{dV}{dt} = \frac{dV_R}{dt} + \frac{1}{C} i$$

DC source so  $\frac{dV}{dt} = 0$  ...  $0 = \frac{dV_R}{dt} + \frac{1}{C} i$

$$R \frac{di}{dt} + \frac{1}{C} i = 0 \Rightarrow \frac{di}{dt} + \frac{1}{RC} i = 0$$

C.F  $\rightarrow$  Transient response (or) source free ckt

$$\frac{di}{dt} + \frac{i}{RC} = 0$$

for RL ckt:  $\frac{di}{dt} + \frac{R}{L} i = 0$

Just  $\frac{R}{L}$  is changed by  $\frac{1}{RC}$ . so

$$i(t) = A \cdot e^{-t/RC}$$

Steady

P.I = steady state response (or)

final value

under Steady state

$$C \rightarrow \infty \Rightarrow i = 0$$

$$i(t) = C.F + P.F$$

$$= A e^{-t/RC} + 0$$

$$\therefore i(t) = A e^{-t/RC}$$

W.K.T  $A = i(0^+) - i(\infty)$

At  $t=0^-$ ;  $V_C = 0 \rightarrow$  since capacitor uncharged.

At  $t=0^+$ ;  $V_C = 0$

$t=\infty$ ;  $V_C = V \Rightarrow \underline{i=0}$ ...

By KVL

$V = V_R + V_C$

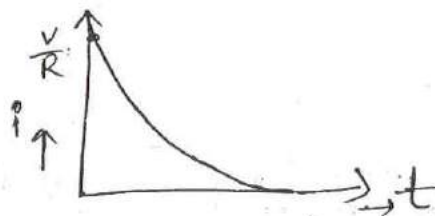
KVL corresponds to  $t=0^+$ ;  $V = V_R + 0 \Rightarrow V = IR \Rightarrow \boxed{I = \frac{V}{R}}$

" "  $t=\infty$ ;  $V = V_R + V_C \Rightarrow V = I \cdot R + V_C$   
 $V = 0 + V_C$   
 $\boxed{V = V_C}$ ;  $i(\infty) = 0$   
 $\therefore \underline{i(0)} = 0$

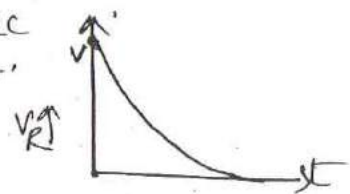
$\therefore A = \frac{V}{R} - 0$

$A = \frac{V}{R}$

$\therefore i(t) = \frac{V}{R} e^{-t/RC}$



$V_R = IR = \underline{\underline{V \cdot e^{-t/RC}}}$



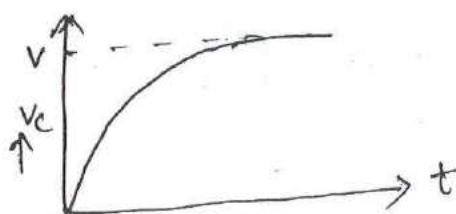
$V_C = \frac{1}{C} \int_0^t i dt$   
 $= \frac{1}{C} \int_0^t \frac{V}{R} e^{-t/RC} \cdot dt$

$V_C(t) = -V \cdot e^{-t/RC} + V$

$V_C(t) = [V_C(0^+) - V_C(\infty)] e^{-t/RC} + V_C(\infty)$

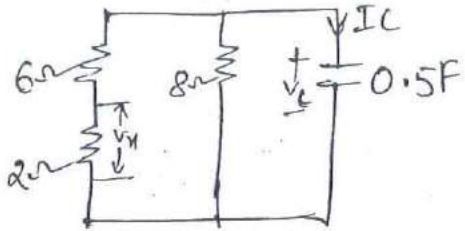
This is the valid general expression for RL & RC CKT. Just  
 $\frac{R}{L}$  &  $RC$  are interchange  
 Initial value  
 final value  
 final value

$V_C(t) = V [1 - e^{-t/RC}]$



- (Pb) find response of  $V_C$ ,  $I_C$  &  $V_X$  when initial value of capacitor is 3V?

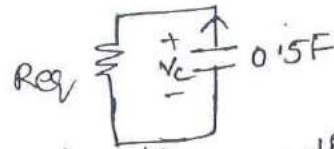
Sol:- RC ckt without source.



$$C_{eq} = 0.5F$$

$$R_{eq} = 4\Omega$$

$$RC = T = 4(0.5) = 2$$



$$V_C = V_0 e^{-t/RC} \rightarrow \text{discharging action}$$

$$V_C = (3) \cdot e^{-t/2}$$

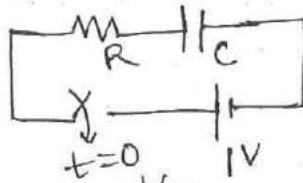
$$V_X = \frac{3}{4} \cdot e^{-t/2}$$

$$I_C = C \frac{dV_C}{dt} = (0.5) \cdot \frac{d}{dt} (3e^{-t/2})$$

$$= -\frac{3}{4} e^{-t/2}$$

(Pb)

find rate of direct voltage across the capacitor at  $t=0^+$ .



- @ RC (C) 0  
(B) RC (D) 1V

Sol:-

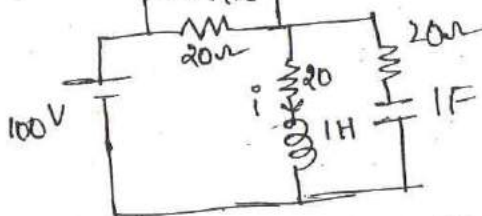
$$V_C = -V e^{-t/RC} + V \rightarrow \text{charging capacitor}$$

$$\frac{dV_C}{dt} = +V \cdot \left(\frac{1}{RC}\right) \cdot e^{-t/RC} + 0$$

$$\frac{dV_C}{dt} = \frac{V \cdot 1}{RC} \cdot e^{-t/RC}$$

$$\text{At } t=0^+ ; \frac{dV_C}{dt} = \frac{V(1)}{RC} = \frac{1}{RC} \quad (\because V=1V)$$

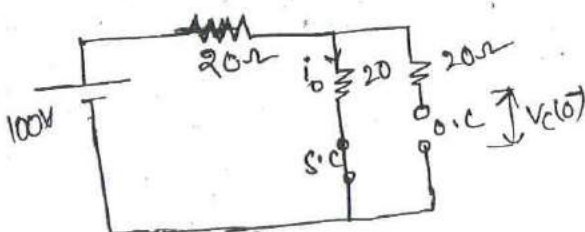
(Pb)



find  $\frac{di}{dt}$  at  $t=0^+$ .

Sol:-

At  $t=0^-$  :- It is a steady state condition.



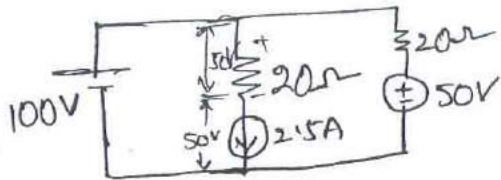
$$i_0 = \frac{100}{40} = 2.5 A = i_L(0^-)$$

$$V_0 = 50V = V_C(0^-)$$

$t=0^-$ : Inductor does not allow sudden change of current, capacitor " " " voltage.

so  $i_L(0^+) = 2.5A$

$V_C(0^+) = 50V$



$V_L = L \frac{di}{dt}$

$50 = L \cdot \frac{di}{dt}$

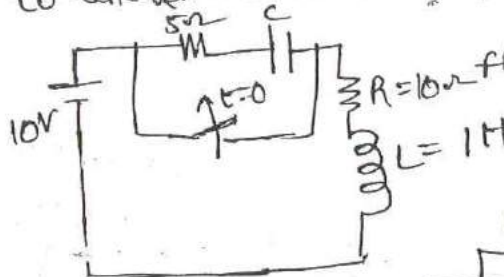
$\frac{50}{1} = \frac{di}{dt}$

$\frac{di}{dt} = 50 \text{ A/sec}$

( $\because L=1H$ )

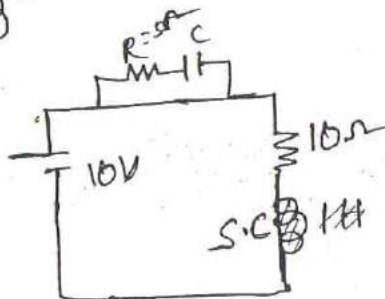
Note:- In some problems initial conditions are already given so we don't need to calculate initial values at  $t=0^-$ .

(P6)



Sol:-

At  $t=0^-$ :



$R_{eq} = 10\Omega$

$X_{eq} = 1H$

$\therefore \frac{R}{L} = \tau = 10 \text{ sec}$

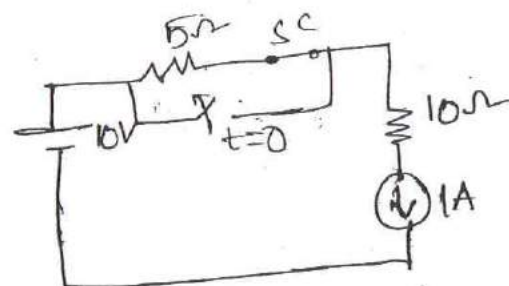
$i_L(0^-) = \frac{V}{R} = \frac{10}{10} = 1A$

$V_C(0^-) = 0V$

$\therefore$  At  $t=0^+$

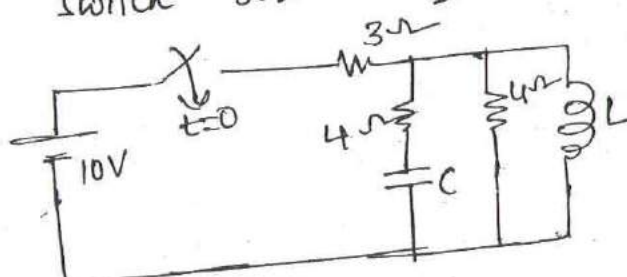
$V_C(0^+) = 0V$

$i_L(0^+) = 1A$



$V_{\text{switch}} = V_{5\Omega} = 5 \times 1 = 5V$

(P6)

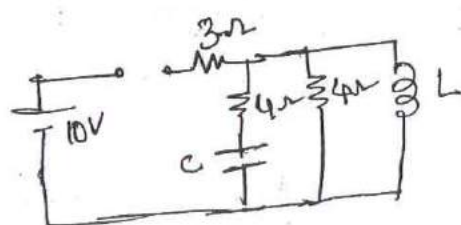


At  $t=0^-$ :

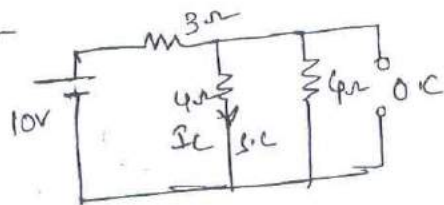
$i_L = 0$

$V_C = 0$

find current flow through capacitor & voltage across inductor at  $t=0^+$ .



At  $t=0^+$ :-



$$I_C(0^+) = \frac{V_{4\Omega}}{4}$$

$$V_{4\Omega} = 10 \times \frac{(4 // 4)}{(4 // 4) + 3} = 10 \times \frac{2}{5} = 4V$$

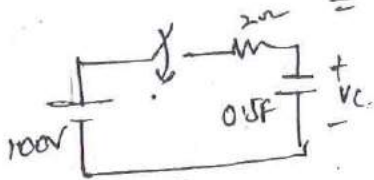
$$\therefore I_C(0^+) = \frac{10 \times \frac{4 \times 4}{4+4}}{4+4} = \frac{10}{5} = 2A$$

$$= \frac{4}{4} = 1A \dots$$

$$V_L = V_{4\Omega} = \frac{4V}{1} = 4V$$

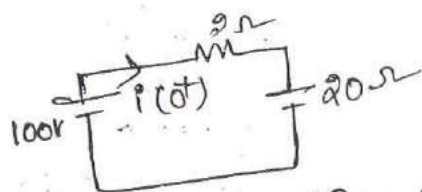
$$= 10 \times \frac{4}{4+4} = 5V$$

(Pb)

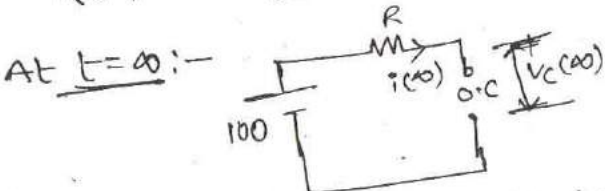


Sol:- At  $t=0^+$  :  $Q_0 = 10C$

$$V_0 = \frac{Q_0}{C} = \frac{10}{0.5} = 20$$



$$i(0^+) = \frac{100-20}{2} = 40A \dots$$



$$i(\infty) = 0$$

$$V_C(\infty) = 100V \dots$$

(i) Energy of the capacitor at  $t=\infty = \frac{1}{2} C (V_C(\infty))^2$

$$= \frac{1}{2} \cdot (0.5) (100)^2$$

$$= 2500J \dots$$

(ii) Current of the capacitor at  $t=\infty$ :-

$$i(t) = [i(0^+) - i(\infty)] e^{-t/RC} + i(\infty)$$

$$= [40 - 0] e^{-t/1} + 0 \quad (\because RC=1)$$

$$i(t) = 40e^{-t}$$

now  $i(1\text{sec}) = 40e^{-1}$

$$= 14.7A \dots$$

(iii)

$$W_R = \int_0^{\infty} P dt = \int_0^{\infty} i^2 \cdot R dt = \int_0^{\infty} (40e^{-t})^2 \times 2 dt$$

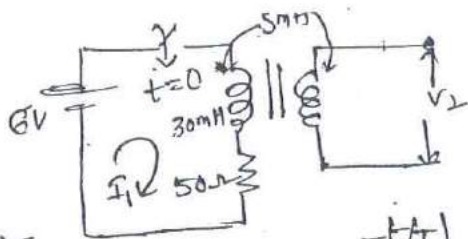
$$= 1600 \left[ -e^{-2t} \right]_0^{\infty} = 1600J \dots$$

In the circuit shown initial charge of the capacitor is 10 Coulombs. Find

- Energy of the capacitor at  $t=\infty$
- Current of the capacitor at  $t=1\text{sec}$
- Energy of the resistor from 0 to  $\infty$  interval.

capacitor doesn't changes sudden changes in voltage & polarities also.

(Pb)



Find max voltage of the ckt shown?

Sol:-

$$I_1(t) = \frac{V}{R}(1 - e^{-\frac{R}{L}t}) = \frac{V}{R}(1 - e^{-\frac{R}{L}t})$$

$$I_1(t) = \frac{6}{50}(1 - e^{-\frac{50}{30 \times 10^{-3}}t})$$

$$V_2 = M \frac{di}{dt}$$

$$V_2 = (5 \times 10^{-3}) \frac{d}{dt} \left( \frac{6}{50} (1 - e^{-\frac{50}{30 \times 10^{-3}}t}) \right)$$

$$= 5 \times 10^{-3} \times \frac{6}{50} \left( 0 + \frac{50}{30 \times 10^{-3}} e^{-\frac{50}{30 \times 10^{-3}}t} \right)$$

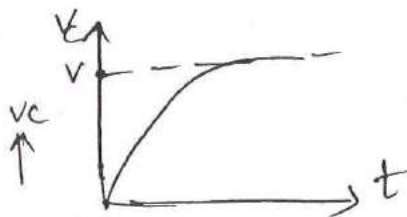
$$= \frac{2}{6 \times 10^{-4}} \left( \frac{50}{30 \times 10^{-3}} \right)$$

$$V_2 = e^{-\frac{50}{30 \times 10^{-3}}t}$$

$V_2 = 1V$  is the max voltage.

Time Constant:-

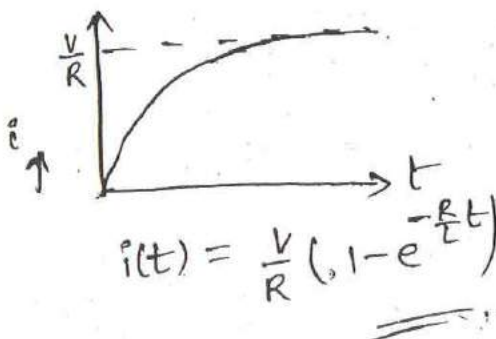
Definition: Time constant can be define either  
1. w.r.to charging (or) discharging  
action of energy storage  
element.



RC circuit with source  
 $V_c(t) = V(1 - e^{-t/RC})$

2. Time constant is the time taken for  
response to rise 63.2% of the  
final value (or) max value, and it  
is given as shown in fig 11

RL circuit with source:-



$$i(t) = \frac{V}{R}(1 - e^{-\frac{R}{L}t})$$

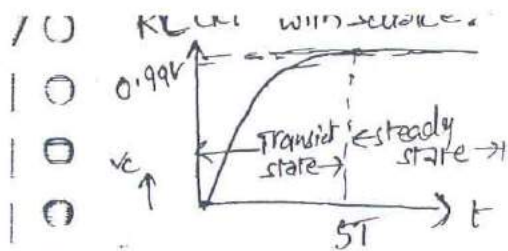
$$V_c(t) = V(1 - e^{-t/\tau})$$

$$t = \tau; V_c(t) = V(1 - e^{-1})$$

$$V_c = 0.632V$$

$$t = 5\tau; V_c(t) = V(1 - e^{-5})$$

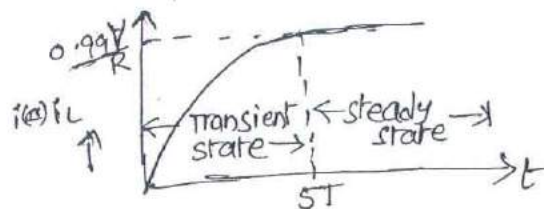
$$V_c = 0.992V$$



$$v_c(t) = V(1 - e^{-t/RC})$$

$$T = RC \text{ see}$$

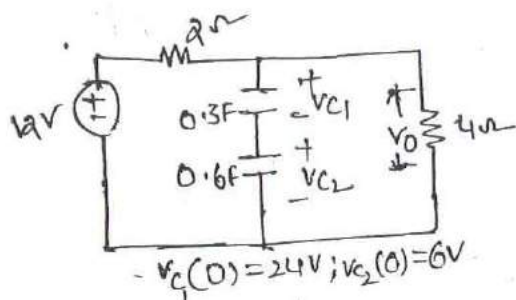
RL CIR with source:-



$$i(t) = \frac{V}{R}(1 - e^{-Rt/L})$$

$$T = \frac{L}{R} \text{ see}$$

(Pb)

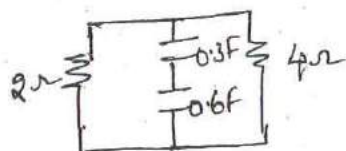


find response of  $V_o$  for  $t > 0$

(a)  $8 + 22e^{-3.75t}$  (b)  $8 + 22e^{-t/0.375}$   
 (c)  $8 + 22e^{-12t}$  (d)  $8 + 22e^{-t/1.2}$

Sol:-

Time constant:-



$$R_{eq} = \frac{2 \times 4}{2+4} = \frac{4}{3}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{0.3 \times 0.6}{0.3 + 0.6} = \frac{3 \times 6}{9} = \frac{1}{5}$$

$$\therefore T = R_{eq} C_{eq}$$

$$\therefore e^{-t/RC} = e^{-3.75t}$$

Ans: (a)

NOTE:- While finding Time constant deactivate all the independent sources.

NOW - General Expression is

$$V_o(t) = [V_o(0^+) - V_o(\infty)] e^{-t/RC} + V_o(\infty)$$

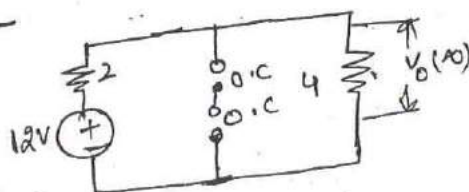
(OP voltage)

$$V_o = V_{C1} + V_{C2} = V_{C1}(0^+) + V_{C2}(0^+)$$

$$= 24 + 6$$

$$= 30V$$

At  $t = \infty$ :-



$$V_o(\infty) = 12 \cdot \frac{4}{6}$$

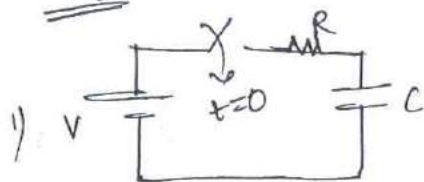
$$= 8V$$

$$V_0(t) = [30 - 8] e^{-t \cdot (3.75)} + 8$$

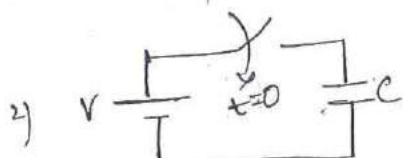
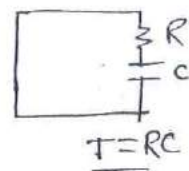
$$V_0(t) = 8 + 22 e^{-8.75t}$$

note: whenever finite time constant is present C & L doesn't allow sudden changes

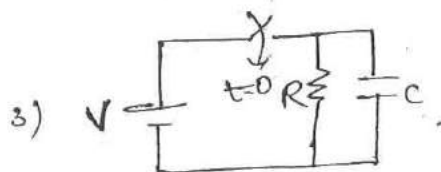
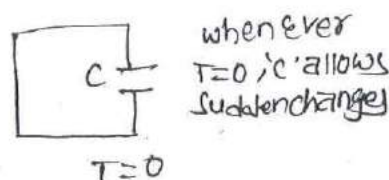
NOTE:-



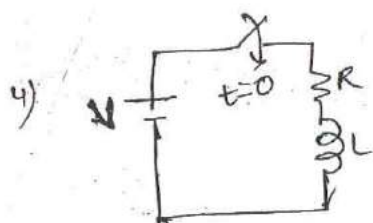
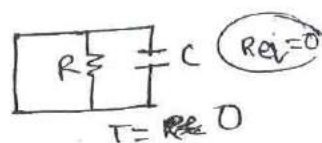
at  $t=0^-$ ;  $V_C = 0$   
 $t=0^+$ ;  $V_C = 0$



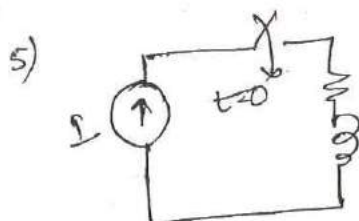
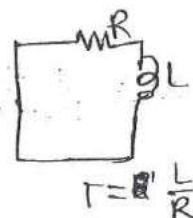
at  $t=0^-$ ;  $V_C = 0$   
 $t=0^+$ ;  $V_C = V$



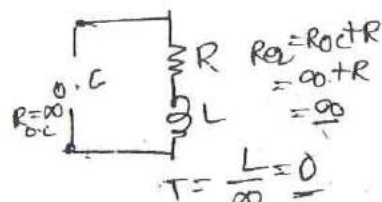
at  $t=0^-$ ;  $V_C = 0$   
 $t=0^+$ ;  $V_C = V$



at  $t=0^-$ ;  $i = 0$   
 $t=0^+$ ;  $i = 0$

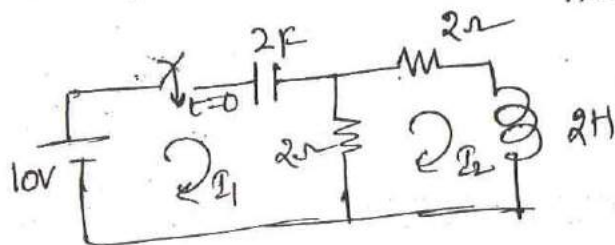


at  $t=0^-$ ;  $i = 0$   
 $t=0^+$ ;  $i = I$



Note:- 2, 3, 5 ckt are practically not exist. Just theoretically because voltage source having some internal impedance (or) resistance which is smaller similarly current source having  $R_{int} = \text{may not } \infty$

(Pb)



find  $i_1(0^+)$ ,  $i_2(0^+)$ ,  
 $\frac{di_1(0^+)}{dt}$ ,  $\frac{di_2(0^+)}{dt}$ ,  $\frac{d^2 i_1(0^+)}{dt^2}$   
 $\frac{d^2 (i_2(0^+))}{dt^2}$  &  $V_C(0^+)$ ?

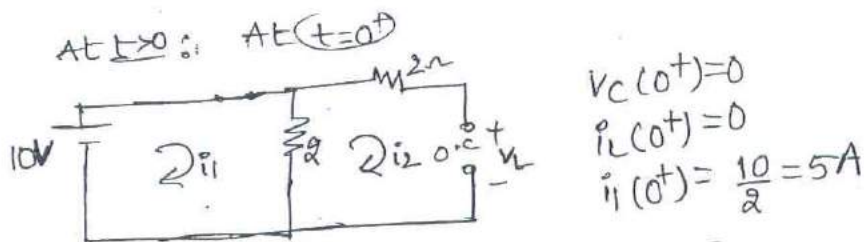
Sol:-

At  $t=0^-$ :-

$$V_C(0^-) = 0 \text{ V}$$

$$i_1(0^-) = 0 \text{ A}$$

$$i_2(0^-) = 0 \text{ A}$$



for loop 2:-  $2i_2 - 2i_1 + 2i_2 + 2\frac{di_2}{dt} = 0 \rightarrow \textcircled{1}$

KVL  $\Rightarrow 4i_2 + 2\frac{di_2}{dt} = 2i_1$

At  $t=0^+$ :-  $4i_2(0^+) + 2\frac{di_2(0^+)}{dt} = 2(i_1(0^+))$

$4(0) + 2\frac{di_2(0^+)}{dt} = 2(5)$

$\frac{di_2(0^+)}{dt} = 5 \text{ A/sec}$

(or)

$V_L = 2(5)$

$L\frac{di_2(0^+)}{dt} = 10$

$\frac{di_2(0^+)}{dt} = \frac{10}{2} = 5 \text{ A/sec}$

At  $t=0^+$ :

KVL:  $-10 + \frac{1}{2} \int i_1 dt + 2(i_1 - i_2) = 0$

$i_1 = i_2$

$\Rightarrow$  Diff w.r. to  $t$

$\frac{i_1}{2} + 2\left[\frac{di_1}{dt} - \frac{di_2}{dt}\right] = 0 \rightarrow \textcircled{2}$

$\frac{i_1(0^+)}{2} + 2\left[\frac{di_1(0^+)}{dt} - \frac{di_2(0^+)}{dt}\right] = 0$

$\frac{5}{2} + 2\left[\frac{di_1(0^+)}{dt} - 5\right] = 0$

$\Rightarrow \frac{di_1(0^+)}{dt} = \frac{1}{2} \cdot \frac{15}{4} \text{ A/sec}$

Diff eqn 1 w.r. to time

$4\frac{di_2}{dt} + 2\frac{d^2i_2}{dt^2} - 2\frac{di_1}{dt} = 0$

At  $t=0^+$ :  $\frac{d^2i_2(0^+)}{dt^2} = \left(2\left(\frac{15}{4}\right) - 4(5)\right)\frac{1}{2}$   
 $= -5.25 \text{ A/sec}^2$

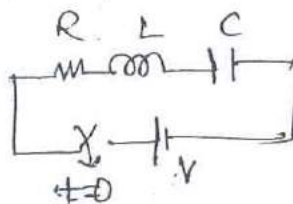
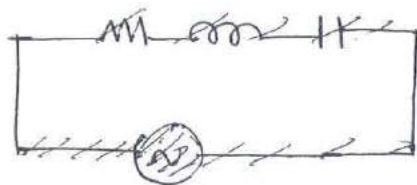
Diff eqn 2 w.r. to time

$\frac{1}{2}\frac{di_1}{dt} + 2\left[\frac{d^2i_1}{dt^2}\right] - 2\frac{d^2i_2}{dt^2} = 0$

At  $t=0^+$ :-  $\frac{1}{2}\frac{di_1(0^+)}{dt} + 2\frac{d^2i_1(0^+)}{dt^2} - 2\frac{d^2i_2(0^+)}{dt^2} = 0$

$\Rightarrow \frac{d^2i_1(0^+)}{dt^2} = -7.13 \text{ A/sec}^2$

## RLC series circuit with DC excitation :-



By KVL :-  $V = iR + L \frac{di}{dt} + \frac{1}{C} \int i dt$

Diff. w.r.to 't'

$$0 = R \frac{di}{dt} + L \frac{d^2 i}{dt^2} + \frac{i}{C}$$

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$$

$$i(D^2 + \frac{R}{L}D + \frac{1}{LC}) = 0$$

Roots :-  $D_1, D_2 = -\frac{R}{L} \pm \sqrt{(\frac{R}{L})^2 - 4 \cdot (\frac{1}{LC})}$

$$D_1, D_2 = -\frac{R}{2L} \pm \sqrt{(\frac{R}{2L})^2 - \frac{1}{LC}}$$

Based on this relation different responses can be get.

Case (i) :-  $(\frac{R}{2L})^2 > \frac{1}{LC}$

This system is called "overdamped system".

$$\alpha = -\frac{R}{2L}$$

$$\beta = \sqrt{(\frac{R}{2L})^2 - \frac{1}{LC}}$$

$$\therefore i(t) = C_1 \cdot e^{(\alpha - \beta)t} + C_2 \cdot e^{(\alpha + \beta)t}$$

Case (ii) :-  $(\frac{R}{2L})^2 = \frac{1}{LC}$

This system is called "critically damped system".

$$i(t) = (C_1 + C_2 t) e^{\alpha t}$$

Case (iii) :-

$$(\frac{R}{2L})^2 < \frac{1}{LC}$$

This system is called "underdamped system".

$$\alpha = -\frac{R}{2L}$$

$$\beta = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

$$\therefore i(t) = (C_1 \cos \beta t + C_2 \sin \beta t) e^{\alpha t}$$

Damping coefficient =  $\frac{R}{2L}$

Time constant =  $\frac{1}{\text{Damping coefficient}}$

$$\text{Time constant for RLC ckt} = \frac{2L}{R}$$

Case (iv) :-

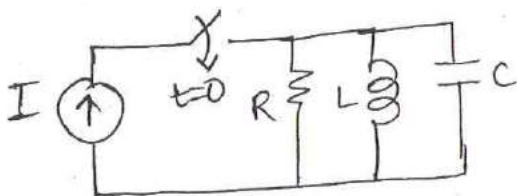
$$R=0$$

undamping

$$\alpha = 0 ; \beta = \frac{1}{\sqrt{LC}}$$

$$i(t) = C_1 \cos \beta t + C_2 \sin \beta t$$

RLC parallel circuit :- (with DC excitation)



By KCL :-

$$I = \frac{V}{R} + C \frac{dV}{dt} + \frac{1}{L} \int V dt$$

$$0 = \frac{1}{R} \frac{dV}{dt} + C \frac{d^2 V}{dt^2} + \frac{1}{L} \cdot V$$

$$\frac{d^2 V}{dt^2} + \frac{1}{RC} \frac{dV}{dt} + \frac{1}{LC} V = 0$$

$$(D^2 + \frac{1}{RC} D + \frac{1}{LC}) V = 0$$

$$D^2 + \frac{1}{RC} D + \frac{1}{LC} = 0$$

$$\text{roots } D_1, D_2 = \frac{-\frac{1}{RC} \pm \sqrt{\left(\frac{1}{RC}\right)^2 - 4\left(\frac{1}{LC}\right)}}{2}$$

$$D_1, D_2 = \frac{-\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}}{2}$$

Case (i) :-

$$\left(\frac{1}{2RC}\right)^2 > \frac{1}{LC}$$

overdamping

$$\alpha = -\frac{1}{2RC} ; \beta = \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

$$V(t) = C_1 e^{(\alpha-\beta)t} + C_2 e^{(\alpha+\beta)t}$$

Case (ii) :-  $\left(\frac{1}{2RC}\right)^2 = \frac{1}{LC}$

critically damping

$$\alpha = -\frac{1}{2RC} = \beta$$

$$V(t) = (C_1 + C_2 t) e^{\alpha t}$$

Case (iii) :-

$$\left(\frac{1}{2RC}\right)^2 < \frac{1}{LC}$$

underdamped system

$$\alpha = -\frac{1}{2RC}$$

$$\beta = \sqrt{\frac{1}{LC} - \left(\frac{1}{2RC}\right)^2}$$

$$V(t) = (C_1 \cos \beta t + C_2 \sin \beta t) e^{\alpha t}$$

Case (iv) :-

$$Q=0 \Rightarrow \frac{1}{R}=0 \Rightarrow R=\infty$$

undamping....

$$\alpha = 0 ; \beta = \frac{1}{\sqrt{LC}}$$

$$V(t) = C_1 \cos \beta t + C_2 \sin \beta t$$

Pb

Sol:-

Find Current Response for  $t > 0$ .  
Steady state Current Response  
of RLC series circuit with DC  
excitation is equal to zero.

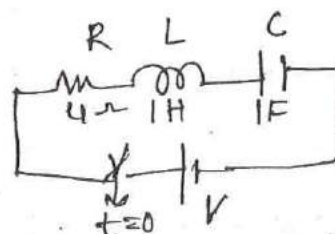
\* series ckt means  
Current response  
\* parallel " voltage  
"

$$\text{damping coefficient} = \frac{1}{2RC}$$

$$\text{Time constant} = \frac{1}{\text{damping coeff}}$$

$$\text{Time Constant} = 2RC$$

for parallel RLC



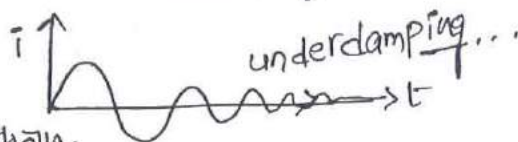
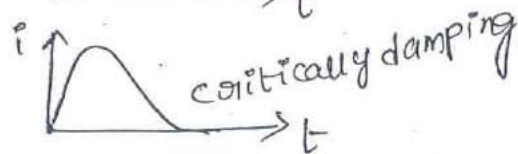
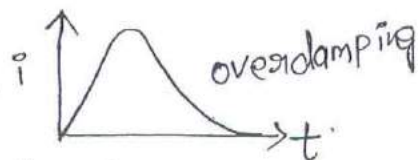
$$\boxed{\tilde{\alpha} = \frac{R}{2} \sqrt{\frac{C}{L}}} \rightarrow \text{for series RLC}$$

$\tilde{\alpha} > 1 \rightarrow \text{overdamping}$

$\tilde{\alpha} = 1 \rightarrow \text{critically}$

$\tilde{\alpha} < 1 \rightarrow \text{underdamped}$

$\tilde{\alpha} = 0 \rightarrow \text{undamped}$



Note:-

\* critically damped system is faster than overdamped system.

\* In the over & critically damped system No oscillations are present.

i.e

$$\tilde{\alpha} \geq 1$$

$$\boxed{\frac{R}{2} \sqrt{\frac{C}{L}} \geq 1}$$

$\rightarrow$  whenever this condition is satisfy No oscillations are produced.

\* for parallel RLC  $\tilde{\alpha} = \frac{1}{2R} \sqrt{\frac{L}{C}} \geq 1$

\* In the under-damped system more than one oscillation is present.

$$\boxed{\frac{R}{2} \sqrt{\frac{C}{L}} < 1}$$

$\rightarrow$  series RLC

$$\boxed{\frac{1}{2R} \sqrt{\frac{L}{C}} < 1}$$

$\rightarrow$  parallel RLC

from the problem

$$R = 4\Omega, L = 1H, C = 1F$$

$$\tilde{\alpha} = \frac{R}{2} \sqrt{\frac{C}{L}} = \frac{4}{2} \sqrt{1} = 2 \geq 1 \dots$$

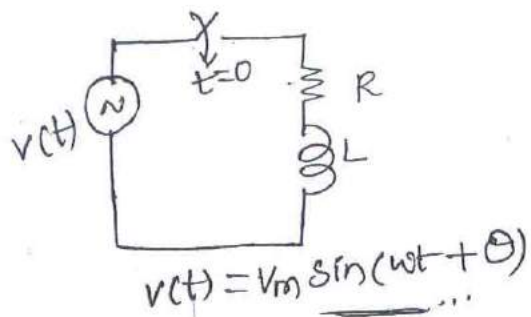
$$\text{So } i(t) = C_1 e^{(\alpha - \beta)t} + C_2 e^{(\alpha + \beta)t}$$

$$\alpha = -\frac{R}{2L} ; \beta = \sqrt{\left(\frac{R}{2L}\right)^2 - \left(\frac{1}{LC}\right)}$$

$$\alpha = -\frac{4}{2} = -2 \dots ; \beta = \sqrt{\left(\frac{4}{2}\right)^2 - \left(\frac{1}{1 \times 1}\right)} = \sqrt{3}$$

$$\Rightarrow i(t) = C_1 e^{(-2 - \sqrt{3})t} + C_2 e^{(-2 + \sqrt{3})t}$$

# 



By KVL:-

$$v = iR + L \frac{di}{dt}$$

$$\Rightarrow \frac{v}{L} = R i + \frac{di}{dt}$$

$$i(t) = \text{C.F} + \text{P.I}$$

(Total Response)

C.F  $\rightarrow$  Complementary function (or)  
(or) Transient response (or) source free response

$$\frac{di}{dt} + \frac{R}{L} i = 0$$

$$i(t) = A e^{-\frac{Rt}{L}}$$

P.I  $\rightarrow$  Steady state Response

$$i = \frac{V}{Z} \angle -\alpha = \frac{V}{Z} \angle -\alpha$$

$$i(t) = \frac{V_m}{Z} \sin(\omega t + \theta - \alpha)$$

$\alpha = \tan^{-1}(\frac{X_L}{R})$   
 $= \tan^{-1}(\frac{\omega L}{R})$

$$i(t) = \text{C.F} + \text{P.I}$$

$$= A e^{-\frac{Rt}{L}} + \frac{V_m}{Z} \sin(\omega t + \theta - \alpha)$$

At  $t=0^-$  ;  $i=0$   
 $t=0^+$  ;  $i=0$

$$0 = A + \frac{V_m}{Z} \sin(\theta - \alpha)$$

$$\Rightarrow A = -\frac{V_m}{Z} \sin(\theta - \alpha)$$

$$i(t) = \underbrace{-\frac{V_m}{Z} \sin(\theta - \alpha) e^{-\frac{R}{L}t}}_{\text{Transient Response (T.R)}} + \underbrace{\frac{V_m}{Z} \sin(\omega t + \theta - \alpha)}_{\text{Steady State Response (S.R)}}$$

Case (i):

$$v(t) = V_m \sin(\omega t + \theta), t=0$$

for Transient free Response at  $t=0 \Rightarrow T.R=0 \Rightarrow \sin(\theta - \alpha) = 0$

$$\theta - \alpha = 0$$

$$\theta = \alpha$$

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) \rightarrow \text{This is the condition for Transient free Response.}$$

Case (ii):

$$v(t) = V_m \cos(\omega t + \theta), t=0 \rightarrow \text{When source is cosine. This is the only diff b/w case (i) & case (ii)}$$

$$\theta = \tan^{-1}\left(\frac{X_L}{R}\right) + \frac{\pi}{2}$$

Note:- for operating the switching it will take some min time i.e.  $t \neq 0$ .  
 → Based on switching action, transient nature varied.

If switch is operated slowly → sparks ↓ } ex: - DPT switch in our lab.  
 " " fastly → " ↑

→ due to switching action only T.R changes, S.R depends on source &  $R, L, C$  etc parameters.

While operating the switch some delay may be present. so

$$i(t) = A e^{-\frac{R}{L}(t-t_0)} + \frac{V_m}{Z} \sin(\omega t + \theta - \alpha)$$

$$\text{At } t=t_0^- : i=0$$

$$t=t_0^+ : i=0$$

$$0 = A e^{-\frac{R}{L}(t_0-t_0)} + \frac{V_m}{Z} \sin(\omega t_0 + \theta - \alpha)$$

$$\Rightarrow A = -\frac{V_m}{Z} \sin(\omega t_0 + \theta - \alpha)$$

Now

Total current is

$$i(t) = \underbrace{-\frac{V_m}{Z} \sin(\omega t_0 + \theta - \alpha)}_{\text{Transient response}} e^{-\frac{R}{L}(t-t_0)} + \underbrace{\frac{V_m}{Z} \sin(\omega t + \theta - \alpha)}_{\text{Steady state response}}$$

Case (3):-

$V = V_m \sin(\omega t + \theta) ; t = t_0$

for Transient Free Response;  $\omega t_0 + \theta - \alpha = 0$

this is the condition for transient free response

$\omega t_0 = \alpha - \theta$

$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta$

Ex:- Application of Transient free system:-

while designing C.B  $\rightarrow$  Arc should be as low as possible.  
i.e. Transient state should be as low as possible. In that cases we are using this concept.

Case (4):-

$V = V_m \cos(\omega t + \theta) ; t = t_0$

$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta + \frac{\pi}{2}$

Note:- RL series circuit conclusions:

Case (i):-  
series  $V(t) = V_m \sin(\omega t + \theta), t = 0$   
parallel  $i(t) = I_m \sin(\omega t + \theta), t = 0$

RL CRT case (i):

$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$   
 $= \tan^{-1}(\omega T)^{**}$

RC CRT

$\theta = \tan^{-1}(\omega RC)$

Case (ii)  $v(t) = V_m \cos(\omega t + \theta), t = 0$

$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) + \frac{\pi}{2}$   
 $= \tan^{-1}(\omega T) + \frac{\pi}{2}$   
\*\*

$\theta = \tan^{-1}(\omega RC) + \frac{\pi}{2}$

Case (iii):  $v(t) = V_m \sin(\omega t + \theta), t = t_0$

$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta$   
 $\omega t_0 = \tan^{-1}(\omega T) - \theta$   
\*\*

$\omega t_0 = \tan^{-1}(\omega RC) - \theta$

Case (iv)  $v(t) = V_m \cos(\omega t + \theta), t = t_0$

$\omega t_0 = \tan^{-1}\left(\frac{\omega L}{R}\right) - \theta + \frac{\pi}{2}$   
 $\omega t_0 = \tan^{-1}(\omega T) - \theta + \frac{\pi}{2}$   
\*\*

$\omega t_0 = \tan^{-1}(\omega RC) - \theta + \frac{\pi}{2}$

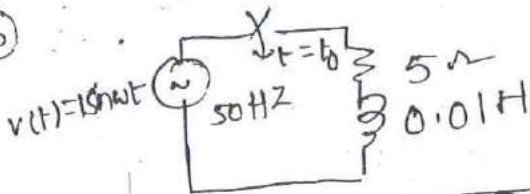
RL KIRCHHOFF'S CIRCUIT

$$i(t) = I_m \sin(\omega t + \theta), t=0 \quad \theta = ?$$

Note:

In RLC circuit it is not possible to obtain transient free response. Since CIR is having two energy storage elements.

(Pb)



In the circuit shown at what value of  $t_0$  transient free response is obtained.

Sol:-

$$\omega t_0 = \tan^{-1} \left( \frac{\omega L}{R} \right) - \theta$$

$\theta = 0$

$$= \tan^{-1} \left( \frac{\omega \cdot 0.01}{5} \right) = \tan^{-1} \left( \frac{\omega(0.01)}{5} \right)$$

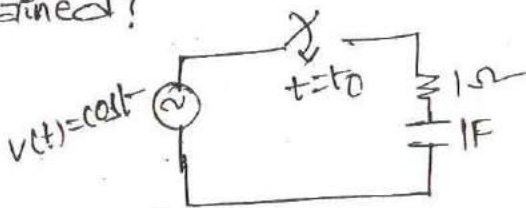
$$\omega t_0 = \tan^{-1} \left( \frac{314(0.01)}{5} \right) = \tan^{-1} \left( \frac{3.14}{5} \right)$$

$$314 t_0 = \tan^{-1} \left( \frac{3.14}{5} \right) \Rightarrow t_0 = \dots \dots \dots 1.78 \text{ m sec}$$

(Pb)

Gate-14  
IES  
10M

In the CIR shown at value of  $t_0$  transient free response is obtained?



Sol:-

$$\omega t_0 = \tan^{-1}(\omega RC) + \frac{\pi}{2} - \theta$$

$$1 \cdot t_0 = \tan^{-1}(1 \cdot 1) + \frac{\pi}{2} - 0$$

$$t_0 = \frac{\pi}{4} + \frac{\pi}{2}$$

$$t_0 = \frac{3\pi}{4} = 3 \times \frac{(22)}{4} = 2.3 \text{ sec}$$

Laplace :-

$$\rightarrow L\{A\} \rightarrow \frac{A}{s}$$

$$\rightarrow L\{e^{at}\} \rightarrow \frac{1}{s-a}$$

$$\rightarrow L\{e^{-at}\} \rightarrow \frac{1}{s+a}$$

$$\rightarrow L\{\sin \omega t\} \rightarrow \frac{\omega}{s^2 + \omega^2}$$

$$\rightarrow L\{\cos \omega t\} \rightarrow \frac{s}{s^2 + \omega^2}$$

$$\rightarrow L\{e^{at} \sin \omega t\} \rightarrow \frac{\omega}{(s-a)^2 + \omega^2}$$

$$\rightarrow L\{e^{-at} \sin \omega t\} \rightarrow \frac{\omega}{(s+a)^2 + \omega^2}$$

$$\rightarrow L\{e^{-at} \cos \omega t\} = \frac{(s+a)}{(s+a)^2 + \omega^2}$$

$$\rightarrow L\left\{\frac{df}{dt}\right\} \rightarrow sF(s) - f(0)$$

$$\rightarrow L\{f+at\} \rightarrow \frac{F(s)}{s}$$

$$\rightarrow L\{f(t)\} \rightarrow \frac{F(s)}{s}$$

$$\rightarrow L\{f(t)\} \rightarrow 1$$

Initial value theorem:-

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s) = \lim_{t \rightarrow 0} f(t)$$

→ valid only for poles > zeros (or) denominator power should be greater than Numerator power.

Final Value theorem:-

$$f(\infty) = \lim_{s \rightarrow 0} sF(s) = \lim_{t \rightarrow \infty} f(t)$$

→ valid only when all the poles should be left half of s-plane.

(Pb) Find initial value of the following functions.

1)  $f(s) = \frac{(s+1)(s+3)}{s(2s+2)(s+4)}$

Sol:-

$$\lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} s \cdot \frac{s(s+1)(s+3)}{s(2s+2)(s+4)}$$

$$= \lim_{s \rightarrow \infty} \frac{s^2(1+\frac{1}{s})(1+\frac{3}{s})}{s^2(2+\frac{2}{s})(\frac{4}{s}+1)}$$

$$= \frac{(1+0)(1+0)}{(2+0)(1+0)} = \frac{1}{2} //$$

2)  $f(s) = 3 + \delta(s)$

Sol:-

$$f(s) = \frac{3}{s} + 1$$

$$\lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} s \cdot \frac{(s+3)}{s}$$

$$\lim_{s \rightarrow \infty} (s+3)$$

Here poles < zeros

∴ Initial value theorem can not be applied.

Since denominator power is less than the Numerator power.

(Pb) find final value for the following function?

①  $F(s) = \frac{(s+1)(s+3)}{s(s+2)(s+4)}$

Sol:-

$$\lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} s \cdot \frac{(s+1)(s+3)}{s(s+2)(s+4)}$$

$$= \frac{(1)(3)}{(2)(4)} = \frac{3}{8} //$$

②  $f(t) = 9 + e^{2t}$

Sol:-  $f(s) = \frac{9}{s} + \frac{1}{(s-2)}$

$$\lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} \frac{9(s-2) + s}{s(s-2)}$$

∴

for the above function final value theorem can not be applied since poles are present in right half of the plane. (unstable system)

Q find final value  $F(s) = \frac{w}{s^2 + w^2}$  (a) 0 (b) 1 (c)  $\infty$  (d) none

sol: Do the inverse Laplace  $\frac{w}{s^2 + w^2} \rightarrow \sin wt$  marginally stable

for marginally stable system also final value theorem can not be applied.

Q current flowing through 4H inductor is given by  $I(s) = \frac{10}{s(s+2)}$  find initial voltage of the inductor?

sol:  $V = L \frac{di}{dt}$

$$I(s) = \frac{10}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} = \frac{5}{s} - \frac{5}{s+2} \Rightarrow 5[1 - e^{-2t}] = i(t)$$

$$\frac{di(t)}{dt} = 5(2e^{-2t})$$

from initial value theorem

$$\lim_{t \rightarrow 0} \left( \frac{di(t)}{dt} \right) = 5[2e^{-2(0)}] = 5(2) = 10$$

$$\therefore V = (4)(10) = 40V$$

(or)  
 $V(s) = sL \cdot I(s)$   
 $= \frac{4 \times 10}{s(s+2)}$   
 $\Rightarrow V(t) = 40e^{-2t}$   
 $V(0) = 40V$

Note:-

$$i_L(t) = \frac{1}{L} \int_{-\infty}^t v(t) \cdot dt$$

$$= \frac{1}{L} \int_{-\infty}^0 v(t) dt + \int_0^t v(t) dt$$

$$= i_L(0^-) + \frac{1}{L} \int_0^+ v(t) dt$$

At  $t=0^+$  :-

$$i_L(0^+) = i_L(0^-) + \frac{1}{L} \int_0^+ v(t) dt$$

Note:-  $i_L(0^+) = i_L(0^-)$   $\rightarrow$  this statement is valid except impulse function. ( $\because v \rightarrow \delta(t)$ )

Explanation:

$$i_L(0^+) = i_L(0^-) + \frac{1}{L} \int_0^+ \delta(t) dt$$

$$= i_L(0^-) + \frac{1}{L} (1)$$

( $\because i_L(0^-) = 0$  for impulse)

$$** i_L(0^+) = \frac{1}{L} \rightarrow (2)$$

Now energy is:  $w_L(0^+) = \frac{1}{2} L i_L^2(0^+)$

$$= \frac{1}{2} L \left( \frac{1}{L} \right)^2 = \frac{1}{2L}$$

$$\therefore w_L(0^+) = \frac{1}{2L} \rightarrow (3)$$

Conclusions:-

from the eqn ① it is concluded that inductor does not allow instantaneous changes for a given input.

From ② & ③ it is concluded that inductor allows instantaneous changes for a given input (impulse input  $v = \delta(t)$ ).

Note:- for capacitor:-

inductor means we concentrate on current eqn, capacitor means we concentrate on voltage equation.

$$v_c(t) = \frac{1}{C} \int_{-\infty}^t i(t) dt$$

$$v_c(t) = \frac{1}{C} \int_{-\infty}^{0^-} i(t) dt + \frac{1}{C} \int_{0^-}^t i(t) dt$$

$$v_c(t) = v_c(0^-) + \frac{1}{C} \int_{0^-}^t i(t) dt$$

At  $t = 0^+$ :-

$$v_c(0^+) = v_c(0^-) + \frac{1}{C} \int_{0^-}^{0^+} i(t) dt \rightarrow \text{since capacitor doesn't allow sudden changes}$$

$$\boxed{v_c(0^+) = v_c(0^-)} \rightarrow ①$$

let us consider impulse function as input.

$$v_c(0^+) = v_c(0^-) + \frac{1}{C} \int_{0^-}^{0^+} i(t) dt$$

$$(\because i = \delta(t))$$

$$v_c(0^+) = v_c(0^-) + \frac{1}{C} \int_{0^-}^{0^+} \delta(t) dt$$

area = 1

$$\boxed{v_c(0^+) = \frac{1}{C}} \rightarrow ②$$

$$w_c(0^+) = \frac{1}{2} C (v_c(0^+))^2$$

$$= \frac{1}{2} C \left(\frac{1}{C}\right)^2$$

$$\boxed{w_c(0^+) = \frac{1}{2C}} \rightarrow ③$$

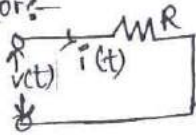
## conclusions :

from eqn ① it is concluded that capacitor does not allow instantaneous changes for given input.

from eqn ② & ③ it is concluded that capacitor allows instantaneous changes for current impulse function.

Transformation of each element from time domain to s-domain :-

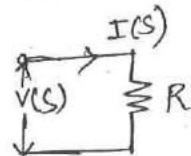
1) Resistor :-



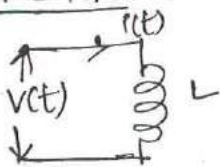
$$v(t) = i(t) R$$

$$v(s) = I(s) R$$

→ since Resistance is independent on frequency.



2) Inductor :-

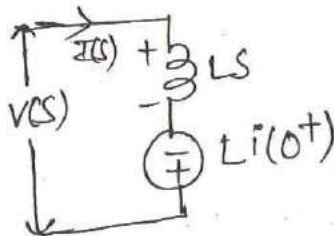


$$v(t) = L \frac{di(t)}{dt}$$

$$v(s) = L [sI(s) - I(0^+)]$$

$$v(s) = LS I(s) - LI(0^+)$$

→ KVL eqn. corresponding to KVL series ckt is drawn since voltage division possible in series only.



$$z(s) = \frac{v(s)}{I(s)}$$

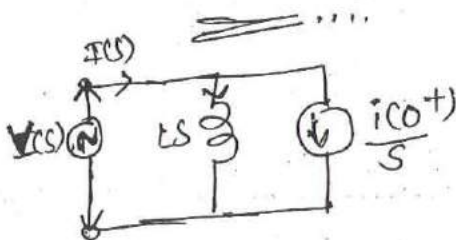
$$z(s) = LS - \frac{LI(0^+)}{I(s)}$$

from ①

$$LS I(s) = v(s) + LI(0^+)$$

$$I(s) = \frac{v(s)}{LS} + \frac{LI(0^+)}{LS}$$

→ current eqn. i.e KCL so parallel ckt is developed.



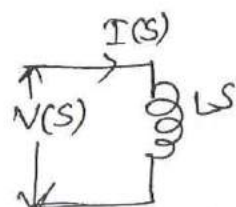
$$y(s) = \frac{I(s)}{v(s)}$$

$$y(s) = \frac{1}{sL} + \frac{I(0^+)}{s v(s)}$$

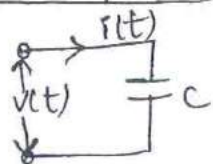
An laplace domine initial value of Inductor is represented as either voltage source (or) Current-source.

if initial value = 0  $\Rightarrow i(0^+) = 0$

Then the equivalent is given by



3) for capacitor :-

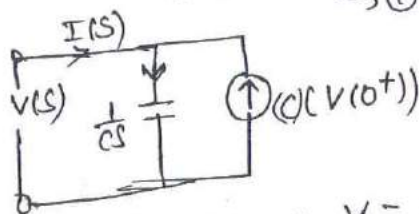


$$i = C \frac{dv}{dt}$$

$$I(s) = C [sV(s) - v(0^+)]$$

$$I(s) = \frac{V(s)}{(1/s)} - (C \cdot v(0^+))$$

→ current eqn i.e KCL so parallel ckt developed.

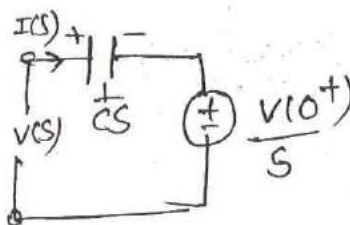
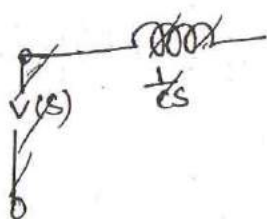


Transformed Admittance  $\Rightarrow Y = \frac{I(s)}{V(s)}$

$$Y = \frac{1}{s} - \frac{C v(0^+)}{V(s)}$$

Now write the voltage value interms of current

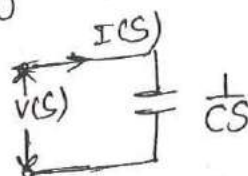
from eqn  $\Rightarrow V(s) = \frac{1}{s} I(s) + \frac{C v(0^+)}{s} \rightarrow$  KVL so corresponds to KVL series ckt developed.



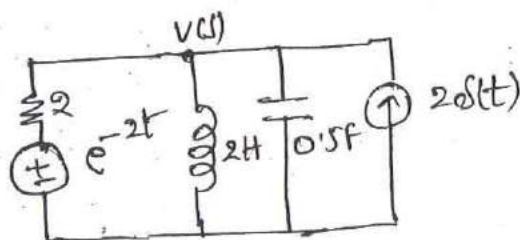
Transformed Impedance  $Z = \frac{V(s)}{I(s)}$

$$Z = \frac{1}{s} + \frac{C v(0^+)}{s I(s)}$$

If  $v(0^+) = 0$



(pb) find  $V(s)$  of the ckt shown?

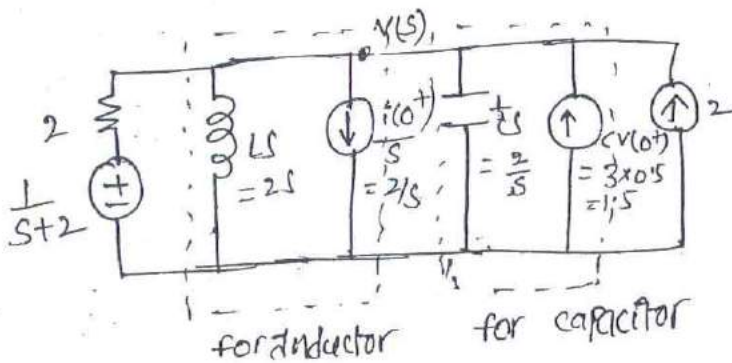


$$i_L(0^+) = 2A$$

$$V_C(0) = 3V$$

So!:-

Initial value of the Inductor & capacitor can be indicated as a Current source (or) voltage source. If all the elements are in parallel then use Current source as Initial values.



concentrate on  
note:- Current directions  
& voltage polarities of  
inductor & capacitor.

Nodal Analysis:-

$$\frac{V(s) - \frac{1}{s+2}}{2} + \frac{V(s)}{2s} + \frac{2}{s} + \frac{V(s)}{(2s)} = 2 + (1.5)$$

$$\Rightarrow V(s) - \frac{1}{s+2} + \frac{V(s)}{s} + \frac{4}{s} + sV(s) = 7$$

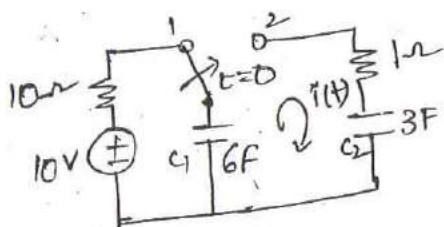
$$\Rightarrow V(s) \left[ 1 + \frac{1}{s} + s \right] = 7 - \frac{4}{s} + \frac{1}{s+2}$$

$$\Rightarrow V(s) = \frac{(7(s)(s+2) - 4(s+2) + s)}{s(s+2)} \times \frac{8}{s+1+s^2}$$

$$V(s) = \frac{(7s^2 + 14s - 4s - 8 + s)}{(s+2)(s^2+1+s)} = \frac{7s^2 + 11s - 8}{s^3 + 3s^2 + 3s + 2}$$

Pb

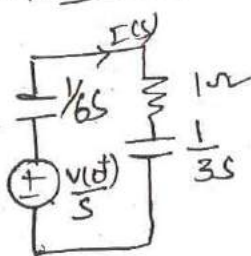
find  $i(t)$  for  $t > 0$



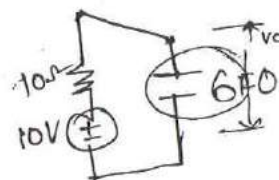
Soln

At  $t=0^-$ :- at  $t=0^-$  at  $t=0^-$  at  $t=0^-$   
steady state. under steady state  
condition capacitor  $\rightarrow 0 \text{ } \Omega$   
so  $V_C = 10V$

At  $t > 0$ :-



corresponds to  $C_1$  initial  
value will be present  
for  $C_2$  nodal value will  
be present. so  $C_1 \rightarrow$



$$V_C = 10V$$

$$\therefore V_C(0^-) = 10V$$

$$V_C(0^-) = 0V$$

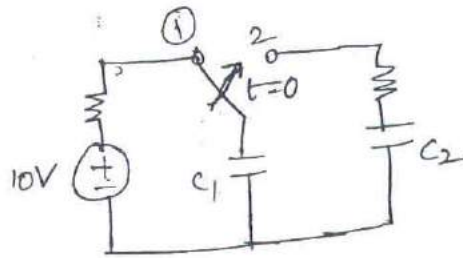
$$C_2 \rightarrow \left[ \begin{array}{l} \text{---} \\ \text{---} \end{array} \right] \text{ s.t. } V(0^-) = 0V$$

$$I(s) = \frac{\frac{10}{s}}{\frac{1}{6s} + 1 + \frac{1}{3s}} = \frac{10(6)}{(1+6s+2)}$$

$$i(t) = \mathcal{L}^{-1}(I(s)) = \mathcal{L}^{-1}\left(\frac{60}{(1+2s)}\right) = 10 e^{-t/2}$$

Verification :-

Classical Method :-



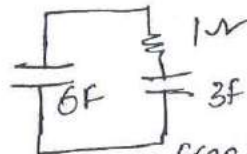
At Position ① :-

$$V_{C1}(0^-) = 10V$$

$$V_{C2}(0^-) = 0V$$

At  $t=0^+$  :-

$$V_0 = V_{C1}(0^-) = 10V$$



$$T = RC = (1) \left( \frac{3 \times 6}{3+6} \right) = \frac{18}{9} = 2$$

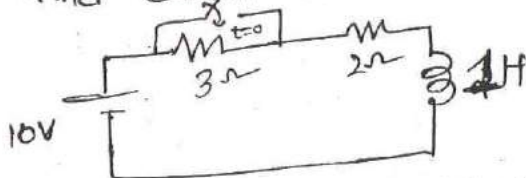
$$V(t) = \frac{V_0}{R} e^{-t/RC}$$

$$i(t) = \frac{10}{1} \cdot (e^{-t/2})$$

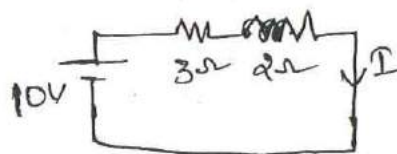
Note :- for conventional exam  $\rightarrow$  Laplace Transform approach is used.  
for objective  $\rightarrow$  classical " " .

(pb)

Find current in the inductor for  $t > 0$ .  
(means complete response have to be determine)

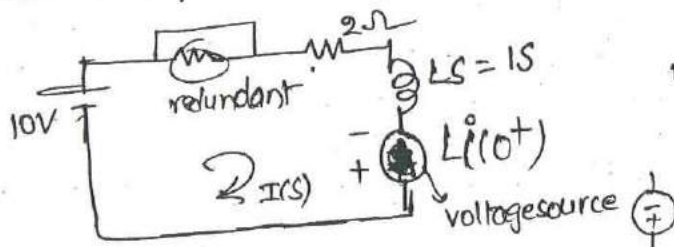


At  $t=0^-$  :- steady state cond<sup>n</sup>  
for DC  $\rightarrow$  inductor S.C



$$I(0^-) = \frac{10}{5} = 2A$$

At  $t=0^+$  (on  $t > 0$  :-



when elements are connected in series take voltage source.

$$I(s) = \frac{\frac{10}{s} + 2}{s+2} = \frac{10+2s}{s(s+2)}$$

$$\Rightarrow \mathcal{L}^{-1} \left( I(s) = \mathcal{L}^{-1} \left( \frac{10}{s(s+2)} + \frac{2}{s+2} \right) \right)$$

$$= 5(1 - e^{-2t}) + 2e^{-2t}$$

Classical Method  
Now Complete Response is given by

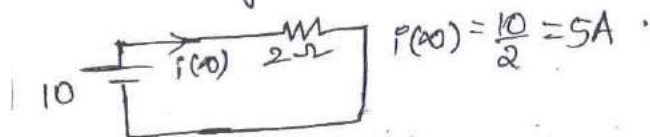
$$i(t) = [i(0^+) - i(\infty)] e^{-\frac{Rt}{L}} + i(\infty)$$

At  $t=0^- = t=0^+ \Rightarrow i = 2A$

At  $t=\infty$ :-

switch closes.

$t=\infty \rightarrow$  Steady state inductor acts as a s.c.



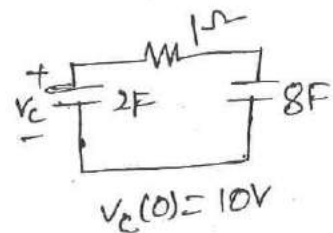
$$i(t) = (2 - 5) e^{-2t} + 5$$

$$= 5 - 3e^{-2t}$$

$\therefore i(0^-) = i(0^+) = 2A$

\*\*\*  
(Pb)

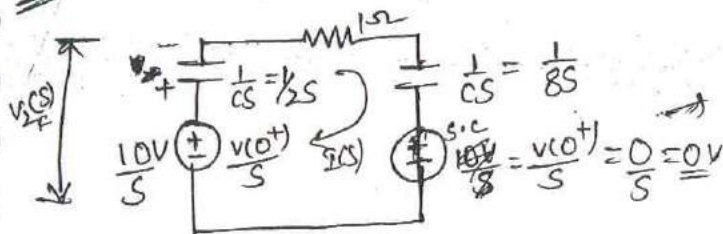
find final value of 2F & 8F capacitor?



Sol:-

let us transform the given ckt into s-domain.

Since  $V_{c(2F)}(0) = 0V$



$$I(s) = \frac{10/s}{1 + \frac{1}{2s} + \frac{1}{8s}}$$

$$V_{8F}(s) = I(s) \cdot \frac{1}{8s}$$

final value of  $V(s)$ :-

$$\lim_{s \rightarrow 0} s V_8(s) = \lim_{s \rightarrow 0} s \cdot \left( \frac{10/s}{1 + \frac{1}{2s} + \frac{1}{8s}} \right) \cdot \frac{1}{8s}$$

$$= 2V \dots$$

$$V_{2F}(s) = \frac{10}{s} - \frac{1}{2s} I(s)$$

$$V_2(s) = \frac{10}{s} - \frac{1}{2s} \left( \frac{10/s}{1 + \frac{1}{2s} + \frac{1}{8s}} \right)$$

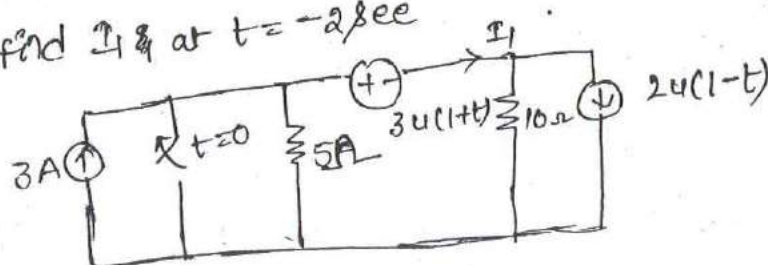
Now final value of  $V_2(s)$ :-

$$\lim_{s \rightarrow 0} s V_2(s) = \lim_{s \rightarrow 0} s \cdot \left[ \frac{10}{s} - \frac{1}{2s} \left( \frac{10/s}{1 + \frac{1}{2s} + \frac{1}{8s}} \right) \right]$$

$$= 8V \dots$$

(Pb)

find  $i_1$  at  $t = -2 \text{ sec}$

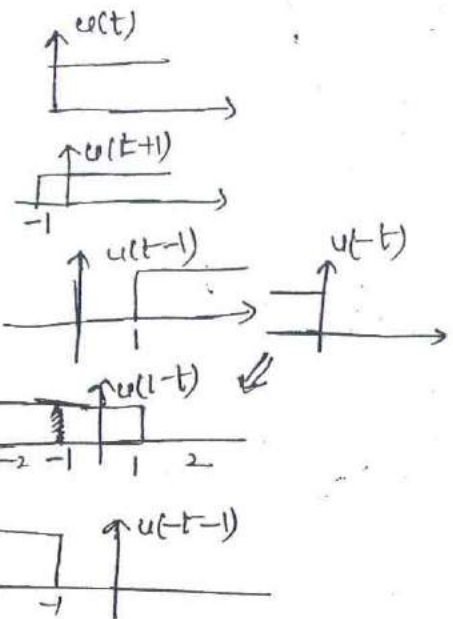
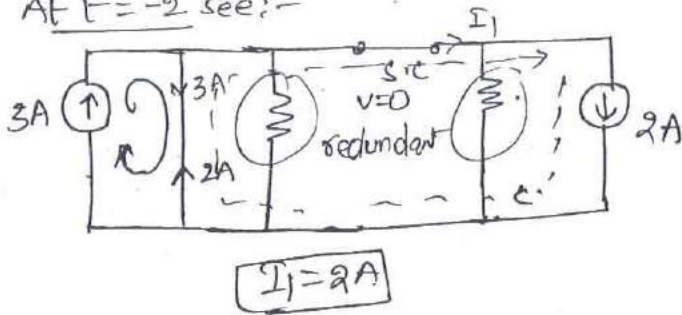


Sol:- For the source:-  $-\infty$  to  $\infty$

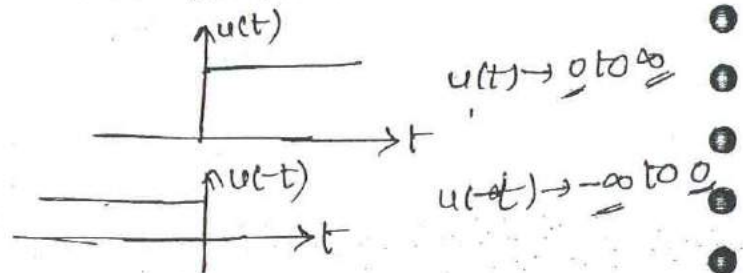
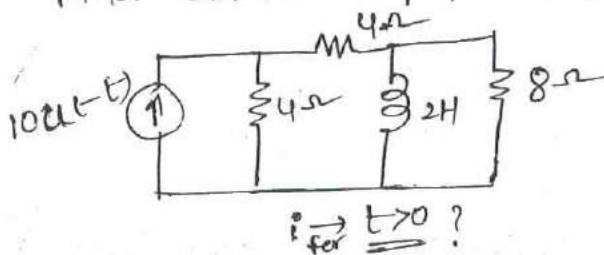
$$u(t) \rightarrow 0 \text{ to } \infty$$

$$u(-t) \rightarrow -\infty \text{ to } 0$$

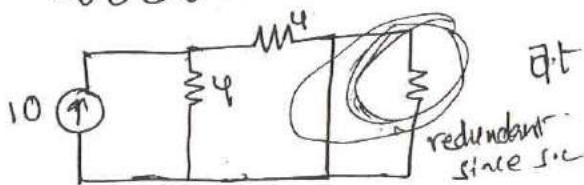
At  $t = -2$  sec:-



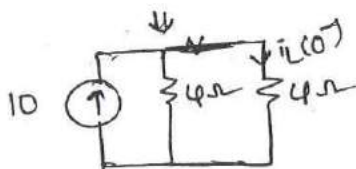
(pb) find current response in the inductor for  $t > 0$ .



Sol:- At  $t = 0^-$ :-

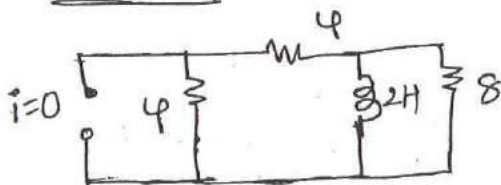


at  $t = 0^- \rightarrow$  steady state condition  
so inductor short ckt



$$i_L(0^-) = 5A$$

At  $t = 0^+$  (or) At  $t > 0$ :-



$$R_{eq} = 4\Omega$$

$$L = 2H$$

$$T = \frac{L}{R} = \frac{2}{4} = \frac{1}{2} \text{ sec}$$

for source free RL ckt:  $i(t) = I_0 e^{-t/\tau}$   
 $= 5 e^{-t/(1/2)}$

$$i(t) = 5 e^{-2t}$$

- Q (10) In the figure shown at  $t=0$  sec the switch is connected to position 1.  
 After 1 Time Constant the switch is transfer to position 2. Find  
 Current response when the switch is at position 2.

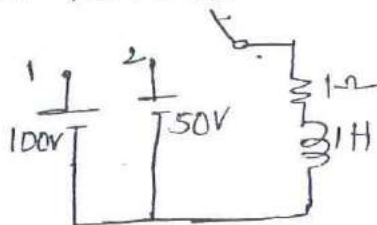
Sol:- At  $t=0^-$ :-

$$i=0$$

At  $t=0^+$ :- (At position 1);  
 Inductor charges  $\Rightarrow$  exponentially rises

$$i(t) = \frac{V}{R} \cdot (1 - e^{-\frac{Rt}{L}})$$

$$i(t) = \frac{100}{1} [1 - e^{-t}]$$



At position 2:-

Here initial values are present. so we consider general formula

$$i(t) = [i(0^+) - i(\infty)] e^{-\frac{Rt}{L}} + i(\infty)$$

This is valid for time  $t=0$  sec.

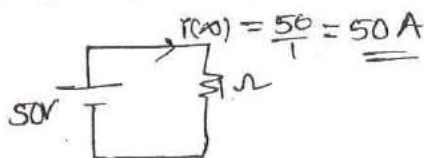
Now the switch is operated at time  $t=1$  sec.

$$\text{So } i(t) = [i(1\text{sec}) - i(\infty)] e^{-\frac{R}{L}(t-t_0)} + i(\infty)$$

$$i(t) = [i(t_0) - i(\infty)] e^{-\frac{R}{L}(t-t_0)} + i(\infty)$$

since  $t=t_0 \Rightarrow i(t_0) = 100 [1 - e^{-1}] = 63.2 \text{ A}$

Now  $t=\infty$  condition  $\rightarrow$  inductor will act as a short ckt.



$$\text{Now } i(t) = [63.2 - 50] e^{-\frac{1}{1}(t-1)} + 50$$

$$= 13.2 e^{-(t-1)} + 50$$

This is the current response corresponds to position 2.

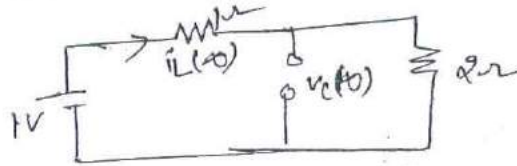
Note:- 'to' delay time will be present since after 1 Time Constant the switch is transferred. So  $t_0=1$  sec.

Workbook

Q:1  $\omega = \frac{1}{24}$  Joules

Q:2  $t=0^-$  |  $t=0^+$   
 $V_C(0^-) = 0$  |  $I_L(0^-) = 0$   
 $V_C(0^+) = 0$  |  $I_L(0^+) = 0$

$t=\infty$  : Inductor  $\rightarrow$  SC  
 Capacitor  $\rightarrow$  OC



$I_L(\infty) = \frac{1}{3}$   
 $V_C(\infty) = \frac{2}{3}V$

Q:3:-

under state state condn :

$V_C(\infty) = V_C(0^-) = 2V$

$V(0^+) = 2V$

Now At  $t=\infty$  :  $V_C(\infty) = 0V \rightarrow$

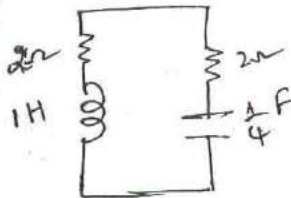


Q:4:

At  $t=0^-$  :  $I_L(0^-) = 1A$

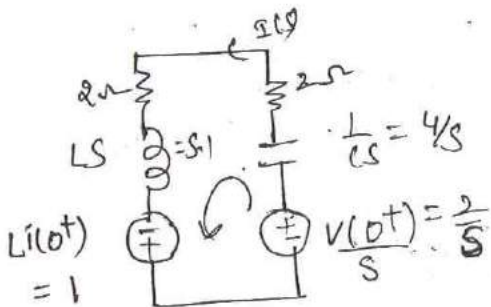
$V_C(0^-) = 2V$

At  $t > 0$  :



Now transform the time domain into 's' domain.

Here all the elements are connected in series.  
 When ever all elements are connected in series indicate the initial condn with initial voltages.



Ans:- b

(or)  
 the above ckt is a series RLC.

so time constant =  $\frac{2L}{R}$   
 $= \frac{2(1)}{4}$   
 $= \frac{1}{2}$

$I(s) = \frac{2s+1}{\frac{4}{s}+2+2+s}$

$= \frac{2+s}{4+4s+s^2} = \frac{s+2}{(s+2)^2}$

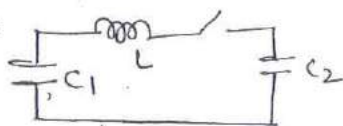
$I(s) = \frac{2}{(s+2)^2} + \frac{s}{(s+2)^2}$

$I(t) = \mathcal{L}^{-1}(I(s))$

$= 2te^{-2t} + e^{-2t}$

$\therefore I(t) = e^{-2t}$

Q 11



Here continuous energy transformation will occur. so oscillatory behaviour will present.

Note :-

In the above ckt energy transformation continuously done b/w inductor and capacitor, thereby oscillatory response is obtained.

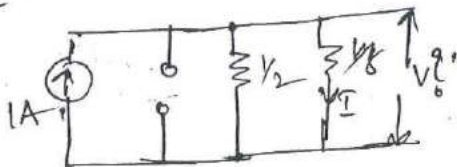
In the above ckt no power loss is present, since resistance is equal to zero.

Q 31

Ans: (a)

Q 5 :-

Sol :-



$$V = \left( \frac{1}{\frac{1}{4} + \frac{1}{2}} \right) \cdot \frac{1}{8} \dots$$

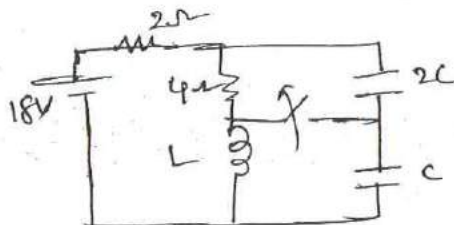
Q 6

Note :-

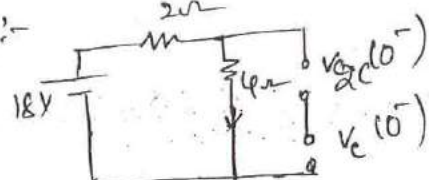
Inductor allows instantaneous changes for voltage impulse function.

Q 17

IES 2012



At  $t=0^-$  :-



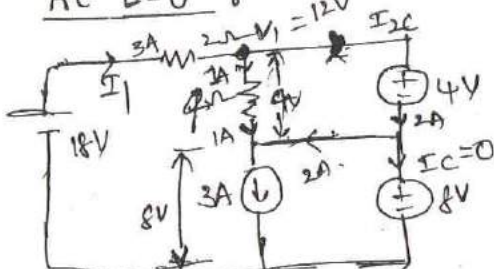
$$I = \frac{18}{6} = 3A$$

$$V = 12V$$

$$\therefore V_{2C}(0^-) = \frac{12 \times C}{C+2C} = \frac{12}{3} = 4V$$

$$\Rightarrow V_C(0^-) = 8V$$

At  $t=0^+$  :-



$$V_1 = 4V + 8V = 12V \text{ (w.r.t to gnd)}$$

$$I_1 = \frac{18-12}{2} = 3A$$

$$I_{2C} = 3 - 1 = 2A$$

$$I_C = 0A$$

Q 8

KVL Eqn

$$V = L \frac{di}{dt} + Ri$$

$$18 = L(6) + 1(6)$$

$$\boxed{L=2}$$

Q 9

$$CP = \frac{1}{P} \times T \cdot F$$

$$C(s) = \left( \frac{1}{s-3} \right) \cdot \left( \frac{1}{s-2} \right)$$

$$= \frac{A}{s-3} + \frac{B}{s-2}$$

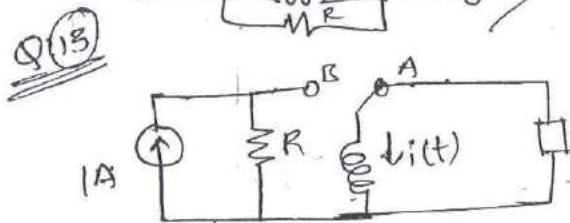
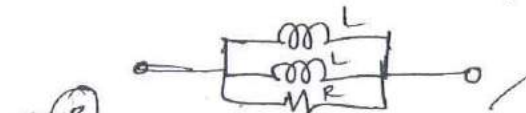
$$1 = A(s-2) + B(s-3)$$

$$C(t) = e^{3t} - e^{2t}$$

$$\begin{aligned} A+B &= 0 \\ B &= -1 \\ A &= 1 \end{aligned}$$

Q: 10) from Initial Value Theorem & Final Value Theorem

Q: 12) At  $t = \infty$ , Total Energy of Inductor will be 100% & so current = 0



when  $S \rightarrow \text{at A}:-$

$$i(t) = I \sin(\omega t + 30^\circ)$$

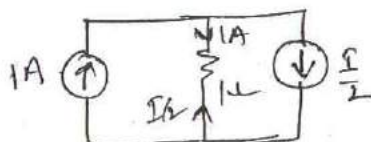
current is leading, so unknown element is capacitor. so A & C are wrong options.

Now At  $t = 0^+:-$

$$i_R = 1A$$

$$i(0^+) = \frac{I}{2} \quad (\because i(0) = I \sin(0 + 30^\circ) = I/2)$$

Now at position (B):-

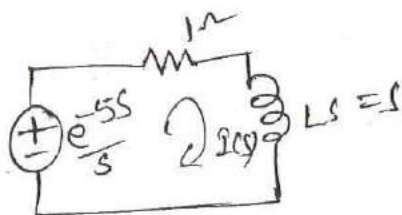


but in the problem they given in both cases power should be constant. so current through the resistor should be same. so

$$-1A + \frac{I}{2} = 1A$$

$$I = 4A$$

Q: 14



transform to s-Domine & then find out inverse Laplace.

Q: 15

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1}{s+1} = \frac{1}{j\omega+1}$$

from given data  $\omega = 1, \dots$

$$= \frac{1}{j(1)+1}$$

$$|M| = \frac{1}{\sqrt{2}} \\ \angle \phi = 45^\circ$$

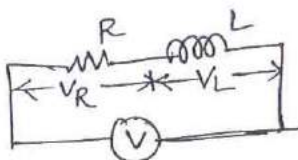
$$\therefore \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{1}{\sqrt{2} \angle 45^\circ} = \frac{1}{\sqrt{2}} \angle -45^\circ$$

$$V_0 = V_i \left( \frac{1}{\sqrt{2}} \angle -45^\circ \right)$$

$$\therefore V_0 = \frac{1}{\sqrt{2}} \cos(t - 45^\circ)$$

Q: 16

Q: 17



So it is a RL with source.  
from given data  $V_R = V_L$   
so  $V_R = V_L = \frac{V}{2}$

now  $V_L = \frac{V}{2}$

$$e^{-t/\tau} = \frac{1}{2}$$

$$\therefore t = -\tau \ln\left(\frac{1}{2}\right)$$

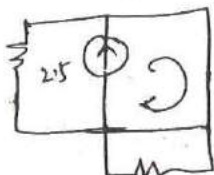
Ans: (b)

Q: 17

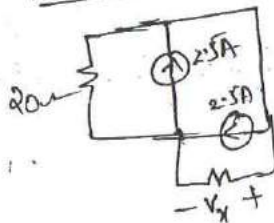
Ans: (c)  
 $i = C \frac{dv}{dt}$   
 $\Rightarrow i = 20 \times 10^{-6} \times (500)$   
 $= 10 \text{ mA}$   
 $50 = iR = \frac{50}{10 \text{ mA}} = 5 \text{ k}\Omega$

Q: 18

At  $t=0^-$ : switch closed



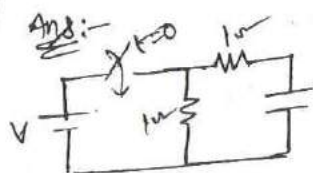
At  $t=0^+$ :



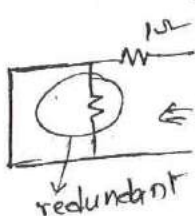
$$V_L = -2.5 \times 20 = -50 \text{ V}$$

NOTE:- while finding time constant deactivate all sources.

Q: 19



So Req:-



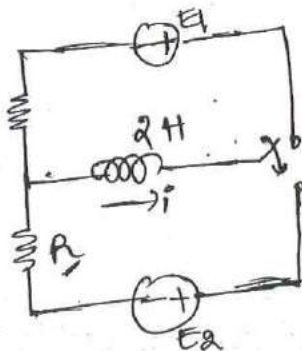
$$R_{th} = 1 \text{ M}\Omega$$

$$C_{eq} = 1 \text{ F}$$

$$\therefore T = RC = 1 \text{ sec}$$

redundant

Q: 20



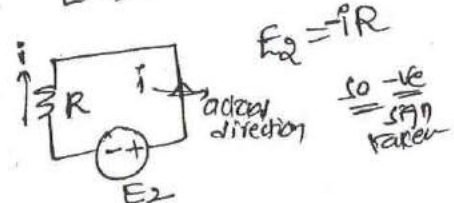
$$i(0) = -8$$

$$\frac{di(0)}{dt} = 3 \text{ A/s}$$

$$i(\infty) = 4 \text{ A}$$

At  $t = \infty$ :

$L \rightarrow \text{SC}$



$$E_2 = iR$$

$$\text{so } \frac{-V_L}{s} = \frac{V_L}{s}$$

$$E_2 = -i(\infty) \cdot R$$

$$= -4R$$

At  $t=0^+$ :

$$Ri + L \frac{di}{dt} + E_2 = 0$$

$$-8R + 2 \cdot (3) + (-4R) = 0$$

$$\Rightarrow 12R = -6 \Rightarrow R = \frac{1}{2}$$

$$\therefore E_2 = -4 \left( \frac{1}{2} \right)$$

$$E_2 = -2 \text{ V}$$

Q: 21

$$v(t) = 10(t - 0.01)e^{-100t}$$

$$V(s) = \frac{10}{(s+100)^2} - \frac{(0.01)10}{s+100}$$

$$= \frac{10 - 0.1(s+100)}{(s+100)^2}$$

$$s^2 + (100)^2 + 200s = 0$$

$$\Rightarrow \omega_n = 100$$

$$2\zeta\omega_n = 200$$

$$2\zeta(100) = 200$$

$$\boxed{\zeta = 1}$$

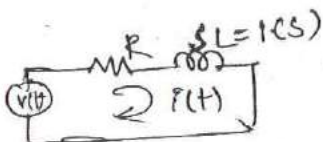
So critically damped system

Q: 23

So:  $v(t) = 10\sqrt{2} \cos(t + 10^\circ) + 10\sqrt{5} \cos(2t + 10^\circ)$

$R = 1\Omega, L = 1H$

$i(t) = ?$



$$I(s) [R + sL] = V(s)$$

$$\frac{I(s)}{V(s)} = \frac{1}{(1+s)}$$

Case (i)

$$= \frac{1}{\sqrt{2}} \angle -45^\circ$$

Case (ii)

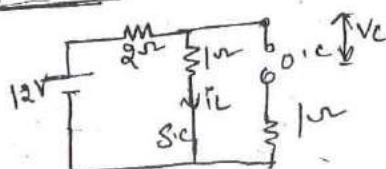
$$= \frac{1}{\sqrt{1+4}} \angle -\tan^{-1}(2)$$

$$= \frac{1}{\sqrt{5}} \angle -\tan^{-1}(2)$$

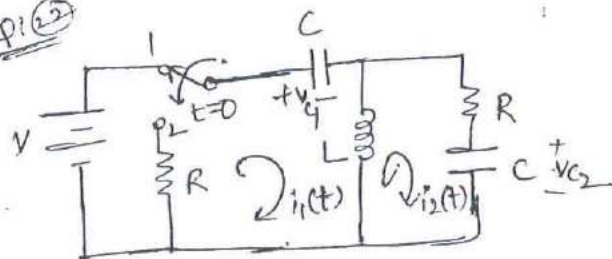
$$\therefore i(t) = \frac{10\sqrt{2}}{\sqrt{2}} \cos(t + 10^\circ - 45^\circ) + \frac{10\sqrt{5}}{\sqrt{5}} \cos(2t + 10^\circ - \tan^{-1}(2))$$

Q: 24

At  $t = 0^-$



Q: 22



At  $t = 0^-$

$$v_C(0^-) = V$$

$$v_L(0^-) = 0$$

$i_L = 0$  since potential diff = 0

Source = V  
Capacitor = V

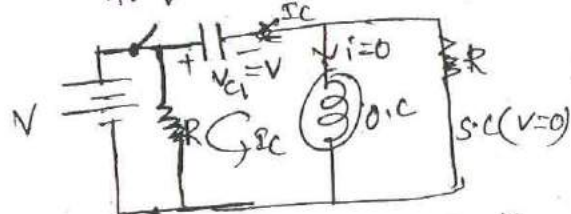
At  $t = 0^+$

$$v_C(0^+) = V$$

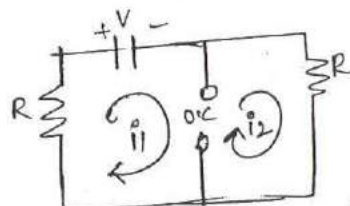
$$v_L(0^+) = 0V$$

$$i_L(0^+) = 0A$$

At  $t = 0^+$  apply KVL



$$i_C = \frac{V}{2R}$$



$$i_1 = -i_C = -\frac{V}{2R}$$

$$i_2 = -i_C = -\frac{V}{2R}$$

$$i_L = i_1 - i_2$$

$$= -\frac{V}{2R} - (-\frac{V}{2R})$$

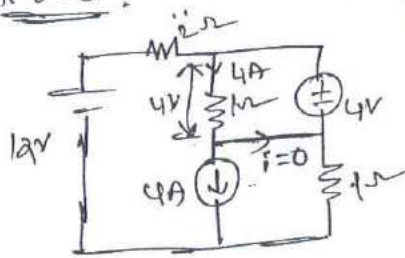
$$= 0A$$

So Ans:  $i_1 = -\frac{V}{2R}$

$$i_L(0^-) = \frac{12}{2+1} = 4A$$

$$v_C(0^-) = 4 \times 1 = 4V$$

At  $t=0^+$ :



$$i_L(0^+) = 4A$$

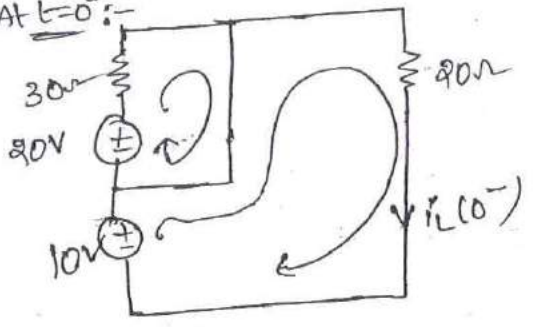
$$v_C(0^+) = 4V$$

Q: 25

complete response is

$$i(t) = [i(0^+) - i(\infty)] e^{-t/\tau} + i(\infty)$$

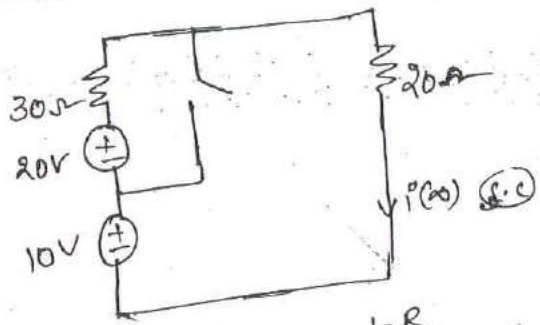
At  $t=0^-$ :



$$i_L(0^-) = \frac{10}{20} = \frac{1}{2} A = 0.5A$$

$$i_L(0^+) = i_L(0^-) = 0.5A$$

At  $t=0^+$ : - switch is open.



$$i(\infty) = \frac{30}{50} = 0.6A$$

$$i(t) = [0.5 - 0.6] e^{-t/\tau} + 0.6$$

$$R = 50\Omega$$

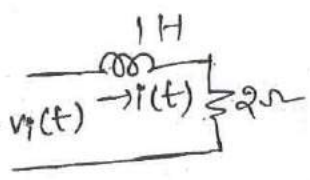
$$L = 1H$$

$$\therefore \frac{L}{R} = \frac{1}{50}$$

$$\therefore i(t) = [0.5 - 0.6] e^{-50t} + 0.6$$

Ans: (B)

Q: 26



$$v_i(t) = u(t)$$

$$\tau = \frac{L}{R} = \frac{1}{2}$$

$$i(s) = \frac{1}{s(2+s)}$$

$$\lim_{s \rightarrow 0} s \cdot i(s) = \frac{1}{s+2}$$

$$= \frac{1}{2} = 0.5$$

$$i(t) = \frac{A}{s} + \frac{B}{s+2} \Rightarrow A(s+2) + Bs = 1$$

$$A = \frac{1}{2}, B = -\frac{1}{2}$$

Note:-

$$i(t) = [i(0^+) - i(\infty)] e^{-t/\tau} + i(\infty)$$

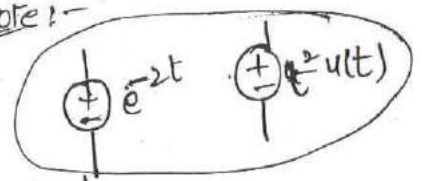
this general expression is valid for

RL  
RC

DC  
(or)  
AC

Not valid for (RLC)

Note:-



When ever source is depends on time, then this general expression is not valid.

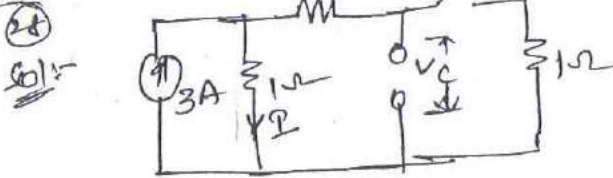
If the source is depends on time we can easy solution by using Laplace transform.

$$i(t) = \frac{1}{2}(1 - e^{-t})$$

$$\text{At } t = \frac{1}{2} \Rightarrow i(t) = 0.31$$

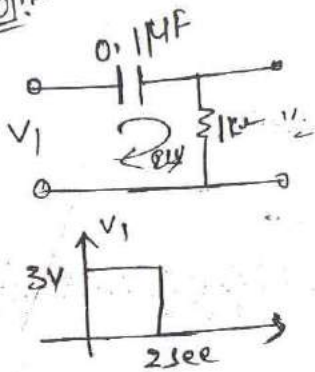
Ans: (c)

Q: 27  
Ans: - b



Q: 29

Sol: -



$$V_1 = 3u(t) - 3u(t-2)$$

$$V_1(s) = \frac{3}{s} - \frac{3e^{-2s}}{s}$$

$$V_2(s) = (1K\Omega) I(s)$$

$$V_1(s) = \left( \frac{1}{0.1 \mu F s} + 1 \right) I(s)$$

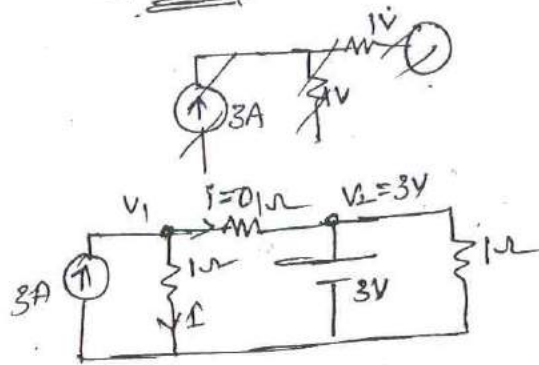
$$\frac{V_2(s)}{V_1(s)} = \frac{1K}{\left( \frac{1}{0.1 \times 10^{-6} s} + 10^3 \right)}$$

$$V_2(s) =$$

$$\text{At } t = 0^- :-$$

$$V_c = 3 \times 1 = 3V ; I(0^-) = 3A$$

$$\text{At } t = 0^+ :- V_c(0^+) = 3V$$



$$\frac{V_1 - 3}{1} + \frac{V_1}{1} = 3$$

$$2V_1 = 6 \Rightarrow V_1 = 3V$$

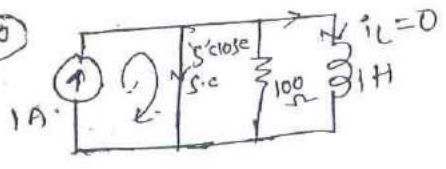
$$\therefore I = \frac{V}{1} = \frac{3}{1} = 3A$$

$$\tau = RC = 10^3 \times 0.1 \times 10^{-6} = 10^{-4} \text{ sec}$$

$$V_2 = -3V$$

In the above ckt at  $t = 2 \text{ sec}$  capacitor discharges immediately its entire voltage to zero. Since time constant of the capacitor is very less.

Q.30



$$v = V_L e^{-\frac{tR}{L}}$$

$$= 100 e^{-\frac{t \cdot 100}{1}}$$

$$= 100 e^{-t(100)}$$

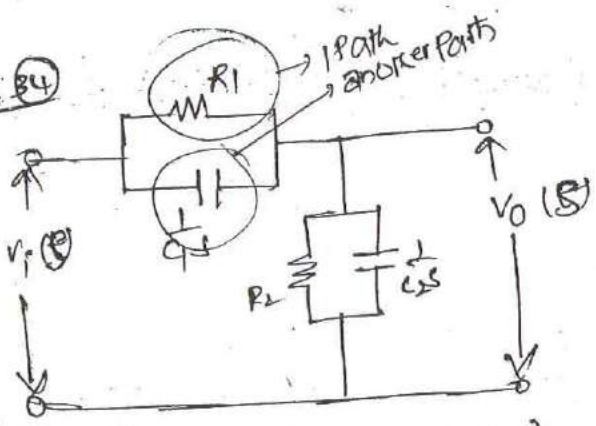
$$\frac{dv}{dt} = -100 \times 100 e^{-100t}$$

$$\left. \frac{dv}{dt} \right|_{t=0} = -100 \times 100 e^{-0}$$

$$= -100 \times 100 \text{ V/sec}$$

Ans: (b)

Q.34



$$V_o = \frac{V_i(s) \left( R_2 \parallel \frac{1}{s} \right)}{\left[ R_1 \parallel \frac{1}{s} \right] + \left[ R_2 \parallel \frac{1}{s} \right]}$$

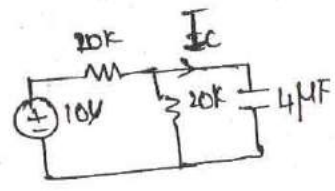
Ans: c

35

$$T = RC$$

$$= (10k)(4nF)$$

$$= 40 \times 10^{-3}$$



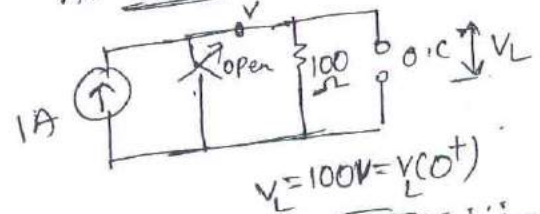
$$i(t) = [i(0^+) - i(\infty)] e^{-t/T} + i(\infty)$$

Ans: (a)

$$i(0^+) = 0.5$$

$$i(\infty) = 0$$

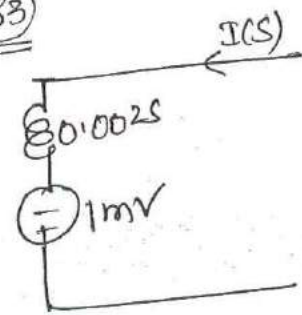
At  $t=0^+$ :  $i_L(0^+) = 0 \text{ A}$



$$V_L = 100V = V_L(0^+)$$

so c & d options are wrong...  
since source is connect, inductor charges.  
so voltage response is decaying  
 $\frac{dv}{dt} \rightarrow -ve$  (slope) so @ option a wrong  
Ans: (b)

Q.33



$$I(0^+) = 1mV$$

$$0.002 I(0^+) = 0.001$$

$$I(0^+) = \frac{1}{2} \text{ A}$$

(a) Time constant of  $\frac{1}{s}$  Path = 0  
"  $\frac{1}{s}$  " = 0

i.e. in the o/c see itself  
'c' reaches to rated voltage  
i.e. o.c (since voltage max at o.c)

$$V_o = V_i \cdot \frac{4k}{1k+4k}$$

$$= 10 u(t) \cdot \frac{4}{5}$$

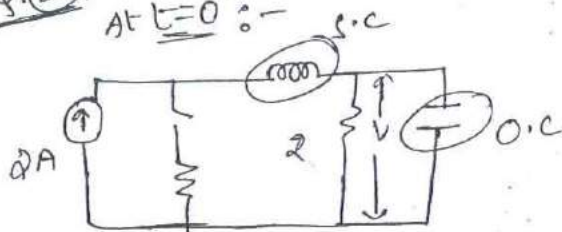
$$= 8 u(t)$$

Notes: in the above circuit time constant of each path of the capacitor is equal to zero

Q: 36) Ans: (c)

Q: 37)

At  $t=0^-$  :-



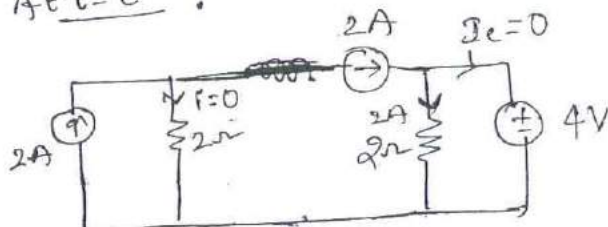
$$I_{2\Omega} = 2A$$

$$\therefore V_{2\Omega} = 4V$$

$$\therefore V_c = 4V = V_c(0^-)$$

$$I_c(0^-) = 2A$$

At  $t=0^+$  :



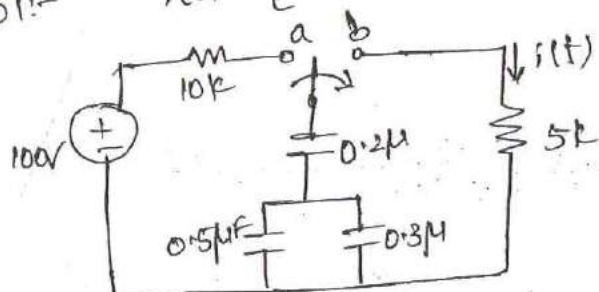
$$I_c = 0$$

$$\frac{dV}{dt} = 0 ; V(0^+) = 4V$$

Q: 38)

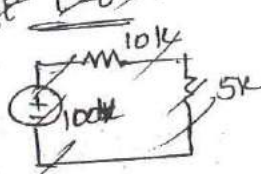
Sol:-

$$i(t) = [i(0^+) - i(\infty)]e^{-t/\tau} + i(\infty)$$



At  $t=0^-$  :-

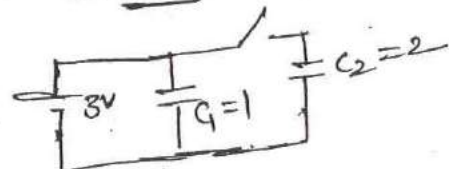
At  $t=0^+$  :-



$$i(0^+) = \frac{100}{15}$$

Q: 39)

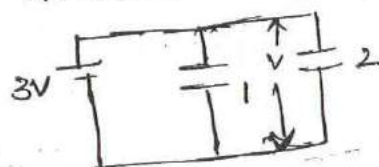
At  $t=0^-$



$$V_c(0^-) = 3V$$

$$\text{charge}(Q) = C_1 V_{C_1} = 1 \times 3 = 3C$$

At  $t=0^+$  :-



Because of no energy loss,  
since no res, total charge  
will be same,

$$Q = 3C$$

$$C_{eq} = C_1 + C_2 = 1 + 2 = 3$$

$$V = \frac{Q}{C} = \frac{3}{3} = 1V$$

Here At  $t=0^-$  :  $V_C = 3V$  } capacitor allows sudden changes bcz  $T=0$   
 At  $t=0^+$  :  $V_C = 1V$  }  $T=RC=0(L)=0$

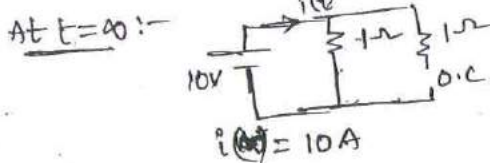
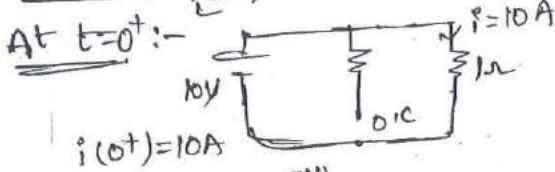
In the above examples capacitor allows sudden change of voltages since time constant of the capacitor  $= 0$ .

Q: 40

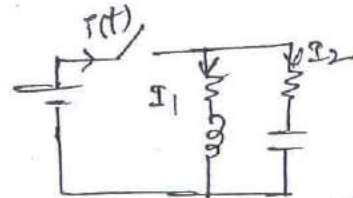
Ans: 6

Q: 41

At  $t=0^-$  :  $i(0^-) = 0$  ;  $V_C(0^-) = 0V$



(Q.3)



$$I_1 = \frac{V}{R} (1 - e^{-\frac{Rt}{L}}) \rightarrow \text{L charging}$$

$$= \frac{10}{1} (1 - e^{-t})$$

$$I_2 = \frac{V}{R} (e^{-t/RC}) \rightarrow \text{C charging}$$

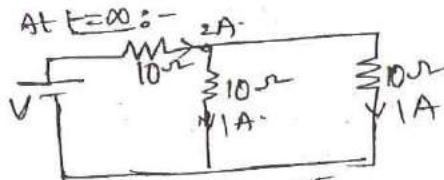
$$= 10(e^{-t})$$

$$\therefore I = I_1 + I_2 = 10A \dots$$

Q: 42 Note :- Time constant should be calculated after operating switch & deactivate all the independent sources.

Ans: - d

Q: 43



$$R_{eq} = 10 + 5 = 15$$

$$\therefore V = I \cdot R_{eq} = (2)(15) = 30V \dots$$

Q: 44 Ans: (c)

$$Y_{12} = \frac{-Z_{12}}{Z_{11}Z_{22} - Z_{12}Z_{21}}$$

$$h_{12} = \frac{V_1}{V_2} \bigg|_{\substack{I_1=0 \\ 0.c}} \quad h_{12} = \frac{Z_{12}}{Z_{22}}$$

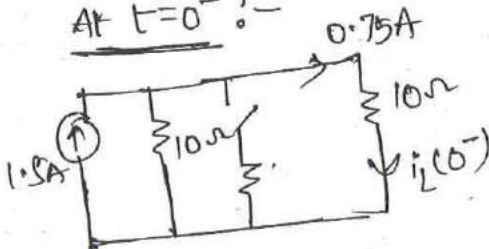
$$A = \frac{V_1}{V_2} \bigg|_{\substack{I_2=0 \\ 0.c}} \quad A = \frac{Z_{11}}{Z_{21}}$$

Q: 45

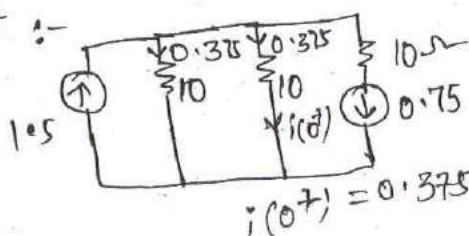
Sol: -

$$i(t) = (i(0^+) - i(\infty)) e^{-t/\tau} + i(\infty)$$

At  $t=0^-$  :-



At  $t=0^+$  :-



$$i(0^+) = 0.375$$

$$\therefore i(t) = \dots$$

Ans: (b)

Q: 46  $T = RC = 10 \times 50 = 500 \mu s$

$$Q = CV$$

$$V = \frac{2.5 \times 10^{-3}}{50 \times 10^{-6}}$$

$$= 50V \dots$$

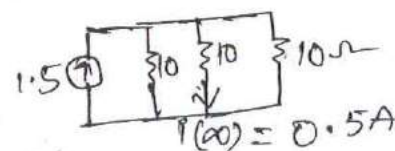
$$i(0^-) = 0V ; V_C(0^-) = 50V$$

$$i(0^+) = 10A \dots$$

$$i(\infty) = 0A$$

$$\therefore i(t) = 10e^{-2 \times 10^3 t} \text{ Amps}$$

At  $t=\infty$  :



Ans: - (d)